

WOOD BADGE PROJECT

Ahtisham Ali Baig

FIRST AID

Proficiency badge



BOY SCOUT SECTION

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Details of certificates

Scout Leader Course: 8-November-2019
Certificate number: SLC/178/19
Wood Badge Training Course: 12-October-2021
Certificate number: SWTC/P1-10./21
Name of councilor: Aly Jaffar Ali
Wood badge Project: First-Aid

GROUP PHOTOS



89th Wood Badge Group Picture



Raven Patrol Photo

Raven Patrol

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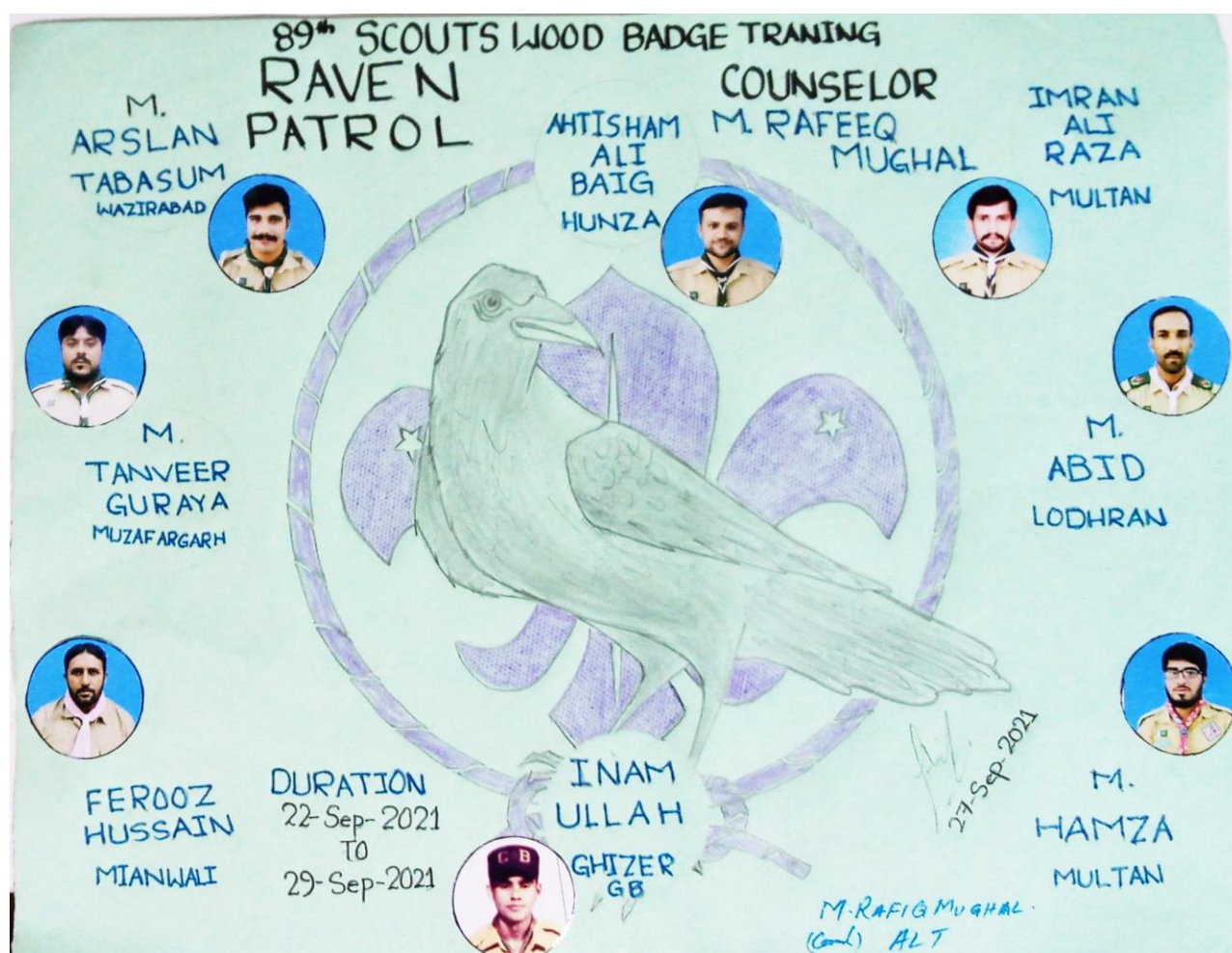


Chart of Raven Patrol with a Patrol Counselor Muhammad Rafiq

Members of Training Team

S.No	Trainers	Qualification	Designation	Contact No
1	Muhammad Adil Farooq	L.T	Course Leader	0312-5846797
2	Muhammad Tariq Qurashi	L.T	Member	0300-5240257
3	Abdul Ghafar Butt	L.T	Member	0334-4397925
4	Muhammad Jamil	L.T	Member	0332-6940433
5	Qazi Muhammad Noor Ullah	L.T	Member	0306-4604886
6	Muhammad Asgar	L.T	Member	0321-9649035
7	Muhammad Rafiq Mughal	A.L.T	Member	0305-4279676
8	Abid Ali Rana	A.L.T	Member	0321-4732241
9	Muhammad Shahbaz	A.L.T	Member	0300-9686054
10	Nabeel Qasim	A.L.T	Member	0303-3252965
11	Hafiz Muhammad Zeeshan	L.T	Course Secretary	0300-6776743

Certificate Scout Leader Course

Punjab Boy Scouts Association

SCOUT LEADER COURSE CERTIFICATE



This is to certify that Mr. AHTISHAM ALI BAIG

S/o DIDAR BAIG of _____

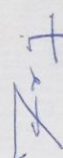
ISMAILI SCOUTS OPEN GROUP, RAWALPINDI has attended

the Scout Leader Training Course held at DIVISIONAL HEADQUARTERS, RAWALPINDI

from 25.10.2019 to 31.10.2019 and qualified as Scout Leader.

No. SLC/178/19

Dated: 08.11.2019


PROVINCIAL SECRETARY
Punjab Boy Scouts Association
Walton, Lahore.

Certificate Scout Wood Badge Training Course Part-I

PUNJAB BOY SCOUTS ASSOCIATION
WALTON - LAHORE



**SCOUT WOOD BADGE TRAINING COURSE
PART - 1**

Certified that Mr. Ahtisham Ali Baig

S/o Didar Baig

Of Ismaili Scouts Open Group, Rawalpindi

has qualified in the Scout Wood Badge Training Course Scout Section Part-1

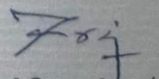
Held at Summer Training Centre, Ghoragali, Murree Hills

From 22-09-2021 **to** 29-09-2021



Certificate No. SWTC/P1- 10./21

Dated 12-10-2021


**Provincial Secretary/
Organising Commissioner**

Topics to be covered in this First Aid book

First Aid is part of Social development of a youth. This book covers in depth all topics required for a standard first aid course, and also includes a section on advanced topics. The basics covered include: from Itihad and Tanzeem then go through in depth of First Aid Proficiency Badge topics.

The topics are as follow:

- A. Definition of first aid and get to know about its objectives and principles
- B. General information on health and Cleanness
- C. Scratches and Skin injury
- D. Uses of triangular and other types of bandages
- E. Epistaxis (Nose bleeding) how to stop it
- F. "Pressure Points" get to know these points and stop bleeding
- G. Handmade stretchers
- H. First aid for Sprain
- I. Burns due to fire and boiling water
- J. First aid on insects and honey bee bites
- K. Covering head, hand, foot, and knee using Triangular Bandage
- L. Preparing sling
- M. Get to know about symptoms and their first aid of following
 - a. Eating poison
 - b. Shock
 - c. Heart Attack
 - d. Choking
 - e. Seizure
- N. Demonstration of CPR (Cardiopulmonary Resuscitation)
- O. First aid on Snake bite and Scorpion Sting
- P. Using only hands for transporting patient
- Q. Getting a first aid knowledge about
 - a. Nervous Pressure
 - b. Mental zest Stress
 - c. Depression
- R. First aid for Heat Stroke, Electric Shock, and in Unconsciousness
- S. Broken bones and its types along its first aid
- T. Respiratory system and two artificially means of giving respiration
- U. Complications due to high blood pressure and low blood pressure
- V. A task for a Boy Scout who is willing to have a First Aid Proficiency Badge

Definition of first aid and get to know about its objectives and principles

What Is First Aid?

Chances are good that at some point in your life, you'll need to provide First Aid to someone who's ill or injured. First aid is the initial care you give to a victim before the arrival of professional medical help, and it can make an important difference in the outcome of an illness or accident—sometimes literally the difference between life and death.

In this First Aid Guide you'll be shown according to the syllabus of Boy Scout Green Book what to do in the case of a victim who is unresponsive, how to treat wounds and broken bones, and what to do if someone is having a heart attack or stroke. The book explains how to treat hypothermia or heatstroke and how to protect against sunburn. You'll also be shown the proper way to administer cardiopulmonary resuscitation (CPR), a lifesaving skill most people associate with First Aid.

This book will show you how to make a quick assessment of a First Aid situation and take measures to stabilize the victim. In any emergency, in addition to the technical help you can offer, an important role is to provide comfort and reassurance and to coordinate calling for professional assistance. Don't underestimate the power of calm, reassuring presence to a victim suffering an emergency.

The book is structured to be a quick reference guide for scouts. Each section gives a basic introduction to the condition or emergency situation, and then describes the common signs and symptoms and first aid treatment. The list of signs and symptoms focuses on the key features of the condition so you can recognize it and offer the appropriate initial First Aid treatment. Illustrations throughout the book provide handy reference guides for you to follow. First Aid does not have to be overly complicated or require in-depth medical knowledge. You never know when you might encounter a first aid emergency, so be prepared! "You are always in a state of readiness in mind and body to do your duty" _ Robert Stephenson Smyth Baden-Powell.

Always Seek Medical Advice

Please remember that this book does not replace professional medical advice from a trained medical professional. You must always seek medical assistance for a victim from a suitable qualified professional without delay. Nor does the book replace a hands-on First Aid and CPR course from an accredited First Aid training provider. After reading this book, consider signing up for a first aid and CPR class in your area so you can have hands-on practice of essential first aid skills. Along scouting many local departments like rescue 1122 offer free CPR training sessions for the general public. If you want more in-depth training for First Aid you can contact your leaders.

Children and babies are at greater risk of accidental death (for example, from choking) remains a leading cause of death among young children. You can prepare yourself for caring for your friends and family by learning how to act in an emergency situation and give your friends and family the best care possible.

Early history and warfare

Skills of what is now known as First Aid have been recorded throughout history, especially in relation to warfare, where the care of both traumatic and medical cases is required in particularly large numbers. The bandaging of battle wounds is shown on Classical Greek pottery from 500 BC, whilst the parable of the Good Samaritan (The British Medical Journal) includes references to binding or dressing wounds. There are numerous references to first aid performed within the Roman army, with a system of First Aid supported by surgeons, field ambulances, and hospitals. Roman legions had the specific role of *capsarii*, who were responsible for First Aid such as bandaging, and are the forerunners of the modern combat medic.

WHAT FIRST AID IS AND ISN'T

FIRST AID is all about providing initial lifesaving care before the arrival of professional help. In First Aid, your main aim is to preserve the life of the victim until they can be treated by Emergency Medical Services **EMS** or another medical professional. A first-aider is someone who has undertaken formal First Aid training but is not a professionally trained emergency worker such as a paramedic, firefighter, or first responder. Reading this book is a good way to learn about basic first aid techniques but does not replace attending an authorized person. In a First Aid situation, you are not expected to act as a paramedic or doctor and perform advanced medical procedures. Don't believe everything you watch in films; you won't be performing any open-heart surgery at the roadside or diagnosing complex medical problems! Instead, you should focus on basic lifesaving interventions to keep the victim alive and stable until EMS arrives to take over. Examples of things you can do in First Aid to save the life of your victim include:

- Opening an unconscious victim's airway and placing him in the recovery position
- Performing cardiopulmonary resuscitation (CPR) and using an automated external defibrillator
- Stopping life-threatening bleeding and recognizing when a victim is going into shock
- Recognizing the signs of a life-threatening medical condition, such as heart attack or insects bites and stings and calling for EMS early
- Cooling a victim who has gone into life-threatening heatstroke

After preserving life, your next aim is to prevent the worsening of the victim's condition. You may not be able to fix the underlying problem affecting the victim. For example, you cannot stop a seizure, but you can prevent worsening of the situation by ensuring the victim's airway is open and protected after the seizure has resolved. In the case

of a suspected neck or back injury, you cannot fix the underlying damage to the spine, but you can prevent the injury from worsening by keeping the victim as still as possible until EMS arrives to take over care.

Finally, your last aim is to promote recovery from the injury or illness. The action you take in the first few minutes of an emergency situation can have a significant impact on the victim's long-term recovery. For example, quickly cooling a major burn will slow down the burning process and reduce the risk of permanent scarring. Another example is performing early effective CPR on a victim of sudden cardiac arrest when the heart has stopped beating properly. Studies have shown that early CPR is associated with a much better chance of the victim making a full recovery from cardiac arrest.

The Overall Aims of First Aid

You can remember your overall aims in first aid by using the three Ps.

They are:

- **P**reserve the life of the victim
- **P**revent worsening of the situation
- **P**romote recovery from the injury or illness

You may recognize the signs of a life-threatening medical condition such as a stroke (brain attack) or heart stroke. These victims require early advanced medical care to have the best chance of recovery. Although you won't be expected to perform this advanced medical care, you can really make a difference by being confident in recognizing the warning signs of these serious conditions and calling for help early.

Symbols

Although commonly associated with first aid, the symbol of a red cross is an official protective symbol of the Red Cross. According to the Geneva Conventions and other international laws, the use of this and similar symbols is reserved for official agencies of the **International Red Cross and Red Crescent**, and as a protective emblem for medical personnel and facilities in combat situations. Use by any other person or organization is illegal, and may lead to prosecution.



ISO First Aid



Emblem of the Red Cross



Emblem of the Red Crescent

Your Role in a First Aid Situation

Calling for Help

In an emergency, the most obvious role you have is to provide the appropriate First Aid for the victim's injury or medical condition. However, you have other roles to carry out in order to manage the situation effectively and provide the best care possible to the victim. Let's take a closer look at some of these responsibilities of a First-Aider.

When an incident occurs, you may be expected to take charge of the situation before the arrival of professional help. The following section, managing an Incident, will walk you through this process. You may have to delegate tasks (for example, calling for help) to other bystanders. You'll need to assume this leadership role during an emergency and take control of the situation prior to emergency responders arriving. People will look to you for guidance, and you may be the only person around with an understanding of first aid. Try to keep calm and provide clear instructions to bystanders. If there are multiple victims, you can instruct bystanders to perform basic First Aid tasks such as applying pressure to a bleeding wound. You also need to ensure that the appropriate professional help has been summoned. Normally, this will involve dialing in Rawalpindi and Islamabad speaking to an emergency operator

Rescue 1122	1122
Emergency helpline:	15
Fire brigade:	16
Edhi Ambulance:	115
PIMS Hospital:	051- 9261223
CDA Hospital:	111-000-232, 9230418
SHIFA Hospital:	051-4603666
Ali Medical Center:	051-2255315
Police:	051-9100008

However, if you are in a remote location and unable to telephone for help, you may need to delegate someone to go and seek assistance. It is vital that you ensure emergency services are called early so the victim receives timely medical treatment. When providing first aid to a victim or victims, recording your actions and any important information is important to enable an effective handover to EMS when they arrive. You may be expected to fill out specific First Aid paperwork depending on the location and severity of the incident. You should take this aspect seriously, although it can seem unimportant when in the middle of a stressful emergency situation. The accurate handover of information to EMS is critical to ensure the victim receives safe ongoing medical care. You also have a responsibility to your health and well-being. Your safety is paramount when dealing with an emergency situation. **Don't put yourself in danger;** you will be unable to help the victim if you are also injured. It can seem unnatural, but you are always the most important person in any situation!

Roles and Responsibilities in First Aid

- Ensure your safety and the safety of bystanders.
- Manage an incident properly and control the situation.
- Call for appropriate emergency services.
- Delegate tasks to bystanders as required.
- Provide appropriate first aid to victims.
- Document your findings and actions.
- Provide an accurate handover to EMS.

Managing an Incident

When an emergency situation occurs, there is often panic among bystanders and victims. In this section, we will look at how to manage an incident and take control of an emergency situation before the arrival of professional help. The priority in managing any incident is to ensure the scene is safe for you and others to approach. You should conduct a quick assessment of any hazards that might be present. A hazard is anything with the potential to cause harm. Make sure you conduct this assessment before rushing in to provide aid to a victim. Otherwise, you run the risk of becoming an additional victim, making you unable to render any aid. If you identify a hazard, then you should take steps to reduce the risk of harm to yourself, bystanders, and the victim. For example, move the victim away from the hazard if possible. If you decide the scene is too dangerous to approach, then stay clear and immediately contact emergency services.

Potential Hazards in a First Aid Situation

- Fast-moving traffic
- Slip and trip hazards
- Extremes of temperature
- Deep or fast-moving water
- Electrical appliances or exposed wires
- The victim, if she or he is under the influence of alcohol or recreational drugs
- Other people (for example, during an altercation)
- Poisons (for example, carbon monoxide gas)
- Exposure to blood or other bodily fluids

General information on health and Cleanness

Good personal hygiene involves keeping all parts of the external body clean and healthy.

It is important for maintaining both physical and mental health.

In people with poor personal hygiene, the body provides an ideal environment for germs to grow, leaving it vulnerable to infection

Types of personal hygiene

Dental

Dental hygiene involves more than just having white teeth. A good dental hygiene routine can help prevent issues such as gum disease and cavities. It can also prevent bad breath

Body

Several million sweat glands cover the human body. When bacteria break down sweat, the process creates a smell or body odor.

Washing the body will help prevent skin irritation, as well as removing the bacteria that cause body odor. Washing the hair removes oil and keeps a person looking clean and fresh.

Hand washing

Regular hand washing is one of the best ways to avoid spreading communicable diseases.

The Centers for Disease Control and Prevention (CDC), recommend washing the hands at certain times:

- Before, during, and after preparing food
- Before eating food
- Before and after looking after anyone who is vomiting or has diarrhea
- Before and after treating a cut or wound
- After going to the bathroom
- After changing diapers or cleaning up a child who has used the toilet
- After blowing the nose, coughing, or sneezing
- After touching garbage or dirty surfaces or objects
- After handling pets or pet-related items, such as food

Nails

Fingernails may harbor dirt and germs, contributing to the spread of bacteria. It is easier for dirt and germs to collect under longer nails, so keeping them short can help reduce the risk of spreading infections

How to maintain good personal hygiene

Knowing how to maintain good personal hygiene can make it easier to build a routine. A person should have some basic knowledge of the following types of hygiene:

Dental hygiene

For a healthy mouth and smile, the Dental Association recommend brushing the teeth for 2 minutes at least twice a day — once before breakfast and once before bed.

People should use an Dental Association-accepted fluoride toothpaste and replace the toothbrush every 3–4 months. The Dental Association also advise people to floss daily.

Hand washing

The Centers for Disease Control and Prevention CDC outline five simple steps for effective hand washing:

1. Wet the hands with clean, running water, then turn off the tap and apply soap.
2. Lather the hands by rubbing them together with the soap, remembering to reach the backs of the hands, between the fingers, and under the nails.
3. Scrub the hands for at least 20 seconds, which a person can time by humming the “Happy Birthday” song twice.
4. Rinse the hands well under clean, running water.
5. Dry the hands using a clean towel or air dry them.

Body

It is advisable to shower or bathe daily, using soap and water to rinse away dead skin cells, oil, and bacteria. People can pay special attention to areas that accumulate more sweat, such as the armpits, in between the toes, and the groin area.

They should also wash their hair with shampoo at least once a week, or more if necessary. Applying deodorant when fully dry can help prevent body odors.

Nails

Using sanitized tools to trim the nails and keep them short is one of the best ways to ensure that no dirt can collect underneath them.

Scrubbing the underside of the nails with a nail brush can form part of a person’s hand washing routine.

Scratches and Skin injury

These guidelines can help you care for minor cuts and scratches:

1. **Wash your hands.** This helps avoid infection.
2. **Stop the bleeding.** Minor cuts and scrapes usually stop bleeding on their own. If needed, apply gentle pressure with a clean bandage or cloth and elevate the wound until bleeding stops.
3. **Clean the wound.** Rinse the wound with water. Keeping the wound under running tap water will reduce the risk of infection. Wash around the wound with soap. But don't get soap in the wound. And don't use hydrogen peroxide or iodine, which can be irritating. Remove any dirt or debris with a tweezers cleaned with alcohol. See a doctor if you can't remove all debris.
4. **Apply an antibiotic or petroleum jelly.** Apply a thin layer of an antibiotic ointment or petroleum jelly to keep the surface moist and help prevent scarring. Certain ingredients in some ointments can cause a mild rash in some people. If a rash appears, stop using the ointment.



5. **Cover the wound.** Apply a bandage, rolled gauze or gauze held in place with paper tape. Covering the wound keeps it clean. If the injury is just a minor scrape or scratch, leave it uncovered.



6. **Change the dressing.** Do this at least once a day or whenever the bandage becomes wet or dirty.

7. **Get a tetanus shot.** Get a tetanus shot if you haven't had one in the past five years and the wound is deep or dirty.

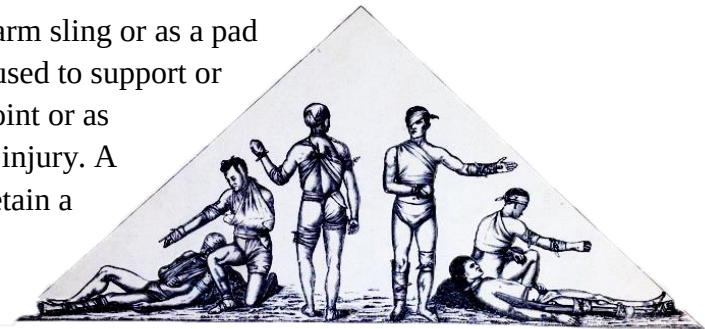
8. **Watch for signs of infection.** See a doctor if you see signs of infection on the skin or near the wound, such as redness, increasing pain, drainage, warmth or swelling.

Uses of triangular and other types of bandages

A triangular bandage is used as an arm sling or as a pad to control bleeding. It may also be used to support or immobilize an injury to a bone or joint or as improvised padding over a painful injury. A tubular gauze bandage is used to retain a dressing on a finger or toe

Triangular bandages

Triangular bandages are usually made from a meter square of cotton or calico that is cut in half diagonally. The bandage can be used in various ways as a sling or for immobilization of broken bones and soft tissue injuries.



Sling

- In the open form as a sling to support an upper body injury.

Broad-fold bandage

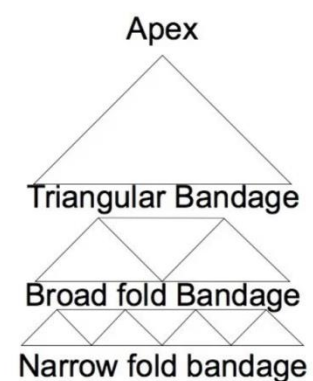
- As a broad-fold bandage with the apex folded down to the base twice to immobilize a lower body injury.

Narrow-fold bandage

- As a narrow-fold bandage with the broad fold bandage folded in half to control severe bleeding, or for immobilization of a lower limb.
- As a collar-and-cuff sling for an upper body injury.

Pad

- As a folded pad after the ends of the narrow fold bandage have been brought into the center three times, and for use on a major wound or as padding.



Tying a reef knot with a triangular bandage

When using a triangular bandage it is important to use a reef knot to secure it in place. A reef knot is a flat knot that will not slip undone and, if correctly placed on the body, is comfortable for the patient.

1. Wrap the left end of the bandage over and then under the right end to start the knot.
2. Wrap the right end over and then under the left end to complete the knot.
3. Pull the knot tightly from both sides to ensure that it will lie flat.

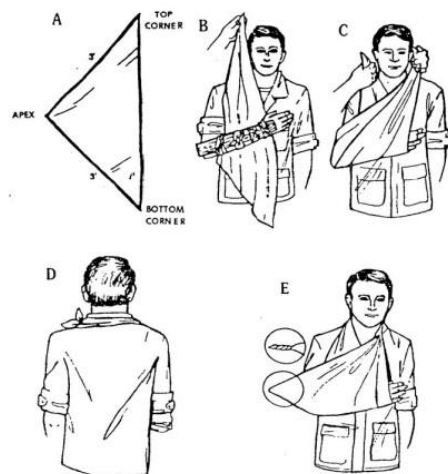
It is easy to untie a reef knot without jarring or hurting the patient. Simply choose two paired ends as they come out of the knot at one side. Then pull the ends apart steadily until two loops form and can be slipped off one end.

Arm sling

This sling is used to support a lower arm or hand

injury and for rib or collarbone fractures.

- Encourage the patient to hold the affected arm across the body in the position of greatest comfort.
- First hold the bandage with the base running down the center of the body and the point to the elbow on the affected side. Gently slip the top point under the supported arm and wrap it around the back of the neck until it rests on the shoulder of the affected side.
- Lift up the lower point and take it to meet the upper point at the side of the neck on the affected side.
- Use a reef knot (see above) to tie the ends together just above the collarbone to avoid any pressure on the back of the neck.
- Adjust the sling so that the fingertips are clearly visible and then bring the point forward and fasten it to the sling with a safety pin.
- Finally, check the circulation in the fingers and compare the tissue color with the fingertips on the unaffected arm. If there are any signs of an impaired circulation, loosen or remove the sling and any underlying bandages.



Elevation sling

This sling is used for an arm or finger injury where the patient needs the hand and arm to be held in an elevated position.

- Encourage the patient to hold the affected arm across the body with the fingers pointing to the opposite shoulder tip.



- First hold the bandage with the base running down the centre of the body and the point to the elbow on the affected side. Gently place the bandage over the supported arm and carry the top end around the front of the neck until it rests on the unaffected shoulder.

- Gently wrap the lower half of the bandage along the affected arm. Carry the free end of the bandage from the elbow across the back to the opposite shoulder tip.



- Gently twist the top point around the fingers, but avoid placing pressure on any injury. Tie the two ends together with a reef knot (see above) and place it just above the collarbone to avoid any neck pressure.

- At the point of the elbow smooth the loose fabric forwards along the arm under the sling. Secure the sling firmly at the elbow with a safety pin or tape.



Collar-and-cuff sling

This sling is used to hold the lower arm and hand in an elevated position where a full elevation sling is either not required, or for patient comfort in very hot weather. The sling



is made with a narrow fold bandage used as a clove hitch.

- Make a clove hitch with two large loops of the bandage. One loop is made with the bandage end pointing upwards and the other end pointing downwards.
- Fold the two loops inwards towards the middle, ensuring that both ends are trapped between the loops.
- Encourage the patient to hold the affected arm across the body with the fingers pointing to the opposite shoulder tip. Then gently slide the two loops over the hand and lower arm with the ends hanging downwards.
- Carry the two bandage ends up on either side of the limb and around the patient's neck. Adjust the bandage so that it is possible to tie a

reef knot (see above) just above the collarbone on one side to avoid any pressure on the neck.

- The knot may be placed on either side of the neck depending on the location of the injury and the comfort of the patient.

Epistaxis (Nose bleeding) how to stop it

Nosebleeds are common. Most often they are a nuisance and not a true medical problem. But they can be both.

Nosebleed care

- **Sit upright and lean forward.** By remaining upright, you reduce blood pressure in the veins of your nose. This discourages further bleeding. Sitting forward will help you avoid swallowing blood, which can irritate your stomach.
- **Gently blow your nose.** Blow your nose to clear your nose of blood clots. Then spray both sides of your nose with a nasal decongestant containing oxymetazoline (Afrin).
- **Pinch your nose.** Use your thumb and index finger to pinch your nostrils shut. Breathe through your mouth. Continue to pinch for 10 to 15 minutes. Pinching sends pressure to the bleeding point on the nasal septum and often stops the flow of blood. If the bleeding continues after 10 to 15 minutes, repeat holding pressure for another 10 to 15 minutes. Avoid peeking at your nose. If the bleeding still continues, seek emergency care.
- **To prevent re-bleeding,** don't pick or blow your nose and don't bend down for several hours. Keep your head higher than the level of your heart. You can also gently apply some petroleum jelly to the inside of your nose using a cotton swab or your finger.
- **If re-bleeding occurs,** go through these steps again. Call your doctor if the bleeding continues.

When to seek emergency care

- The bleeding lasts for more than 30 minutes
- You feel faint or lightheaded
- The nosebleed follows an accident, a fall or an injury to your head, including a punch in the face that may have broken your nose

When to contact your doctor

- **You experience frequent nosebleeds.** You may need to have a blood vessel cauterized. Cautery is a technique in which the blood vessel is burned with an electric current, silver nitrate or a laser. Your doctor may pack your nose with special gauze or an inflatable latex balloon to put pressure on the blood vessel and stop the bleeding.
- **You're experiencing nasal bleeding and taking blood thinners,** such as aspirin or warfarin (Jantoven). Your doctor may advise adjusting your medication dosage.

Apply a water-based lubricant to your nostrils and increase the humidity in your home to help relieve nasal bleeding.



"Pressure Points" get to know these points and stop bleeding

The medical term for bleeding is hemorrhage.

First Aid Treatment for a Major Bleed

1. Immediately call EMS.
2. Wear disposable gloves and apply firm, continuous, direct pressure over the bleeding point.
3. If the wound is on a limb, elevate the limb above the level of the heart to reduce blood loss.
4. If available, apply a sterile pressure bandage directly over the wound
5. If direct pressure does not control the bleeding from a limb, or the bleeding is catastrophic, apply a tourniquet above the wound to control bleeding
6. Monitor for the development of shock, record vital signs, and provide reassurance until EMS arrives.



The direct pressure needs to be applied over the bleeding point of the wound to be effective. Be aware that a large wound may have more than one active bleeding point.

How to apply a direct pressure

Worksite emergencies can happen quickly, and everyone should know what to do - everyone should know First Aid.

Note: You may want to demonstrate the steps for controlling bleeding on a volunteer.

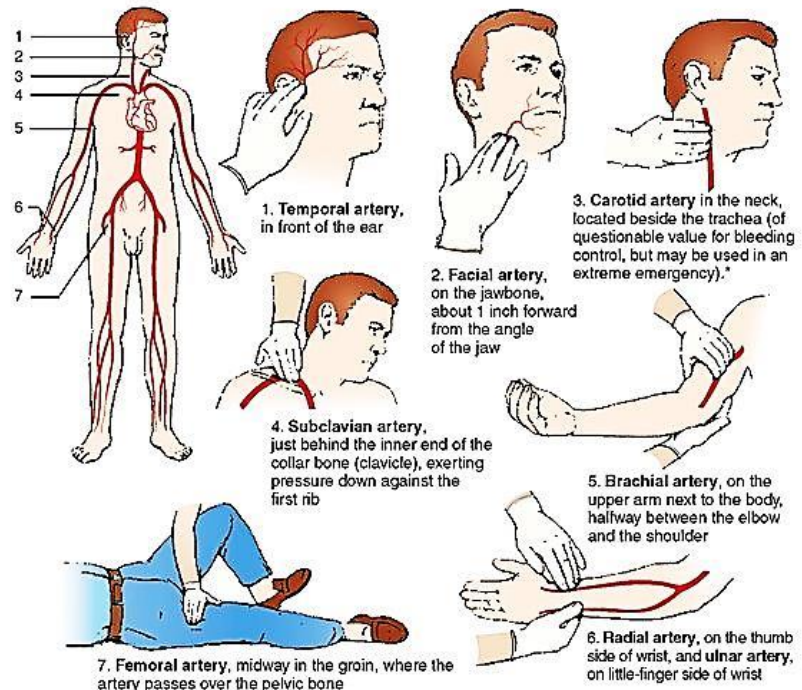
To care for a major open wound (bleeding freely or severely), you must act at once. Begin by applying direct pressure over the wound with a clean, sterile dressing to slow the flow of blood and promote clotting. If a sterile dressing is not available, use any clean cloth such as a towel, handkerchief, or shirt.

Avoid using your bare hands. To reduce the risk of infection or disease transmission, put a barrier between you and the victim's blood, such as disposable gloves or a layer of plastic wrap.

Next, if you do not suspect that the wound involves a broken bone, elevate the injured area above the level of the heart. Elevating the wound will help to slow the flow of blood.



tight, but not so tight that it restricts all circulation beyond the bandage.



*Note: Do not apply pressure to both sides of the neck at the same time. This would cut off the blood supply to the brain.

Remember to maintain direct pressure on the wound at all times, or it may begin to bleed again. If the victim is able to help, have him or her apply the pressure. Use a roll of gauze bandage to wrap snugly around the dressing to keep pressure on the wound. Wrap the bandage around the injured body part, using overlapping turns, covering the entire dressing. The bandage should be

If blood soaks through the bandage, put on more dressing and bandages. Do not remove blood soaked ones. **If the bleeding cannot be controlled, apply pressure to a nearby artery, called a pressure point.** A pressure point is a spot on the body where you can squeeze the artery against the bone. This can slow the flow of blood to the wound. There are two major pressure points in the body. If the bleeding is from the leg, press with the heel of one hand on the **femoral artery** in the groin - where the leg bends at the hip. If the bleeding is from the arm, squeeze the **brachial artery** located on the inside of the upper arm.

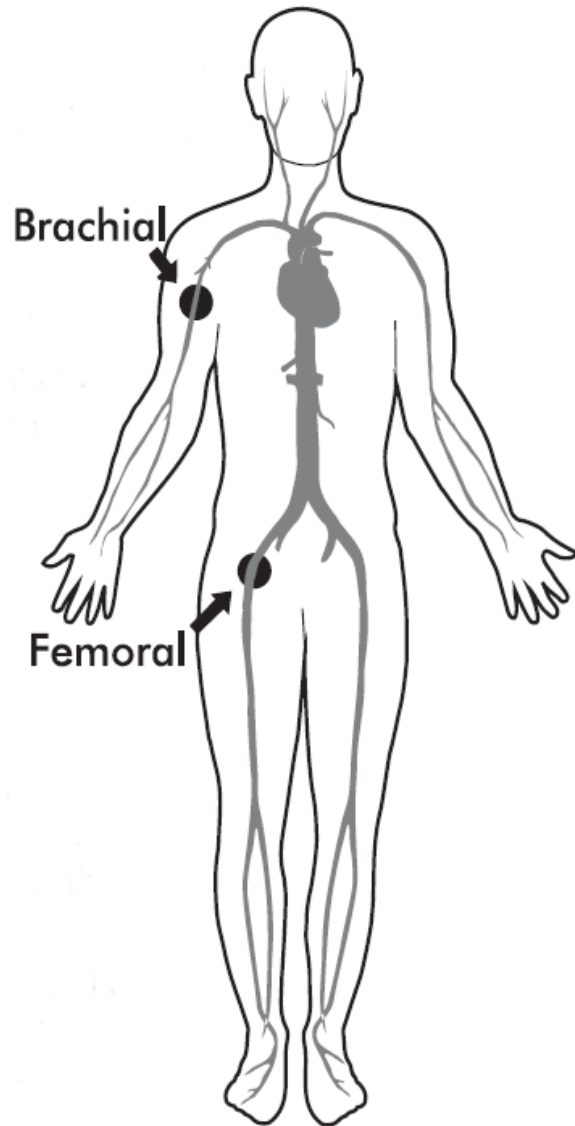
Condition called **shock**. Shock is a condition in which the circulatory systems fails to deliver blood to all parts of the body or irritable, experience rapid breathing or pulse rate, have pale, cool, or moist skin, or drift in and out of consciousness.

To care for a victim of shock:

- have them lie down or rest comfortably;
- elevate the legs about twelve inches if you do not suspect a head, neck or back injury;
- maintain a normal body temperature by covering with a blanket, but do not overheat;
- do not give them anything to eat or drink, even though they are likely to be thirsty;
- call your local emergency number immediately – a victim of shock requires advanced medical care as soon as possible. Call for professional medical help quickly with any severe bleeding emergency. Always wash your hands immediately after completing care, disinfect any blood-contaminated surfaces, and dispose of blood-soaked bandages properly. **Never apply a tourniquet** unless you have been medically trained to do so.

To review, the steps for controlling severe bleeding are:

- Apply direct pressure;
- elevate the wound;
- apply a pressure bandage;
- use a pressure point; and
- treat for shock.



Handmade stretchers

Rule 1: Don't move an injured patient, unless his life is in immediate danger. You can do more damage moving them, so in most cases wait for the paramedics. If, however, the injured person is in a dangerous place like near a fire, rising water or bad weather or dangerous animals are approaching you absolutely have to move an injured person, here's a way to make a quick sturdy stretcher using stuff that's easily available.

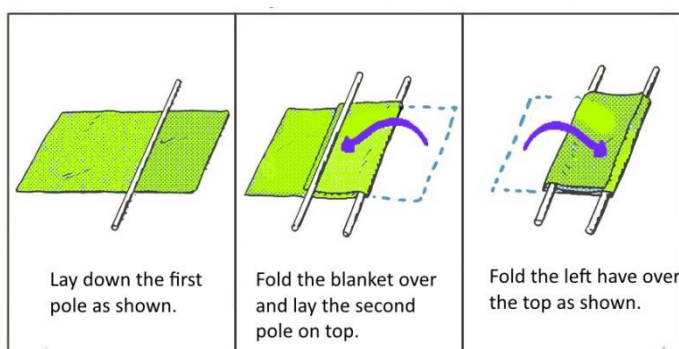
Tarp/Blanket Stretcher

Gather up these two items.

1. A blanket, tarpaulin, sheet of plastic or something blanket-like. On a beach you could use a couple of large beach towels, a sail from a sailboard or some pull the cover off the umbrella.
2. A couple of poles, two tree limbs or long pipes or rods to use for the handles. Two sailboard masts could provide your poles or the umbrella posts. Anything long and straight can work.

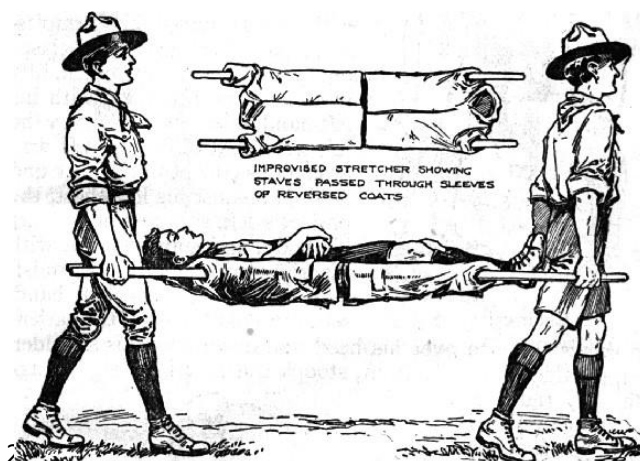
Making the Stretcher

1. Lay out the blanket on a flat place so that the long edge of the blanket is toward you with the short edges to your left and right.
2. Lay one pole on top of the blanket about 1/3 of the way over from the right edge of the blanket.
3. Fold the right flap over the top of the first pole and lay it across the blanket toward the left side.
4. Lay the second pole on top of the left edge of the blanket that was folded over about a third of the way from the left edge of the blanket.
5. Fold the left flap over the top of the pole and lay it all the way across the top of both poles.



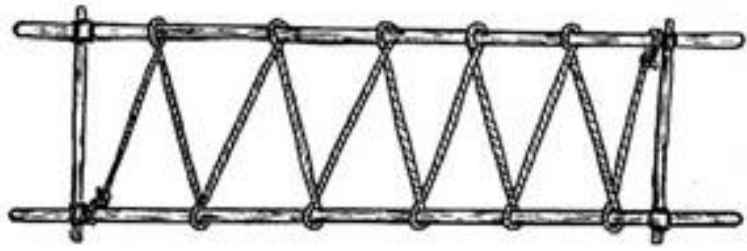
Jacket Stretcher

1. Invert the sleeves of 2-3 jackets (depending on the size of your victim and what's available) so that they run along the inside of the jacket.
2. Thread your poles through the jacket sleeves.
3. Use diagonal lashings to attach cross members at the end of the stretcher to keep the jacket taut and add stability.



Making the Stretcher

1. Take a two bamboo sticks and tie them with a square lashing with a 2 to 2.5ft long sticks on both ends and make it like a rectangle approximately the size of the patient
2. Tie a rope at one end and
3. Now take a long rope and run it through with zigzag pattern.
4. Tie a rope at other end
5. Make it taut, so on lose part remains and it is ready to use.



Moving the Patient:

The best way to move the patient is carefully. Here are the steps.

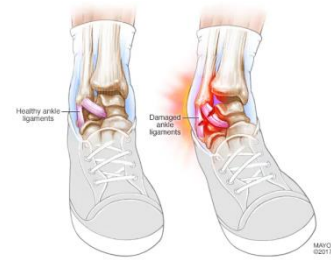
1. Lay the improvised stretcher next to the injured person, parallel to their body.
2. Get as many people as possible to help you lift. The more hands you have helping you, the easier it is to move the patient's body as a single unit.
3. Be careful not to allow damaged body parts to move. I could cause further injury if you try to straighten a broken limb or bend a damaged spine.
4. Lift the patient straight up and move his body slowly and lower it onto the top of the blanket. The patient's weight on top of the blankets will hold the blankets in place and prevent them from slipping from the poles.
5. Put two people on either end of the stretcher or four people, two on each end and lift by the poles.
6. When you carry the stretcher away, make sure everyone carrying it walks out of step with each other. This prevents the stretcher from swinging. If you walk in a march step, the rhythm of your step will cause the stretcher to swing back and forth and can cause those carrying it to stumble or drop the handle.
7. If you have four people carrying the stretcher, put one on each end holding the poles and the other two walking alongside holding the poles at the center.

Note: Before you lift and go, practice a few times quickly so everyone knows what they are going to do. It's a good idea, if you work with kids and/or in wooded or isolated places, to practice making one of these stretchers so you can do it quickly.

First aid for Sprain

Sprain

A sprain is a stretching or tearing of ligaments — the tough bands of fibrous tissue that connect two bones together in your joints. The most common location for a sprain is in your ankle



Symptoms

Signs and symptoms will vary, depending on the severity of the injury, and may include:

- Pain
- Swelling
- Bruising
- Limited ability to move the affected joint
- Hearing or feeling a "pop" in your joint at the time of injury



When to See the Doctor

Mild sprains can be treated at home. But the injuries that cause sprains can also cause serious injuries, such as fractures. You should see a doctor if you:

- Can't move or bear weight on the affected joint
- Have pain directly over the bones of an injured joint
- Have numbness in any part of the injured area

CAUSES

A sprain occurs when you overextend or tear a ligament while severely stressing a joint. Sprains often occur in the following circumstances:

- Ankle — Walking or exercising on an uneven surface, landing awkwardly from a jump
- Knee — Pivoting during an athletic activity
- Wrist — Landing on an outstretched hand during a fall
- Thumb — Skiing injury or overextension when playing racquet sports, such as tennis

KEY POINTS

Children have areas of softer tissue, called growth plates, near the ends of their bones. The ligaments around a joint are often stronger than these growth plates, so children are more likely to experience a fracture than a sprain

RISK FACTORS

Factors contributing to sprains include:

- **Environmental conditions.** Slippery or uneven surfaces can make you more prone to injury.
- **Fatigue.** Tired muscles are less likely to provide good support for your joints. When you're tired, you're also more likely to succumb to forces that could stress a joint.

- **Poor equipment.** Ill-fitting or poorly maintained footwear or other sporting equipment can contribute to your risk of a sprain.

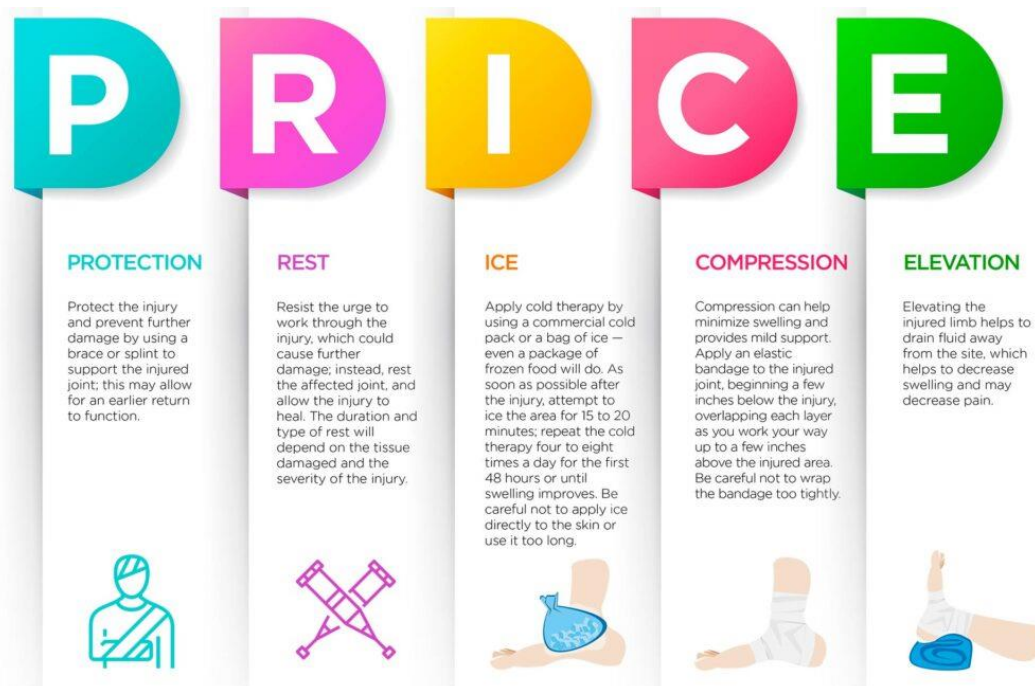
PREVENTION

- Regular stretching and strengthening exercises for your sport, fitness or work activity, as part of an overall physical conditioning program, can help to minimize your risk of sprains. Try to be in shape to play your sport; don't play your sport to get in shape. If you have a physically demanding occupation, regular conditioning can help prevent injuries.
- You can protect your joints in the long term by working to strengthen and condition the muscles around the joint that has been injured. The best brace you can give yourself is your own "muscle brace." Ask your doctor about appropriate conditioning and stability exercises. Also, use footwear that offers support and protection

FIRST AID FOR SPRAIN

Always remember

P.R.I.C.E



The recommended treatment for most acute injuries is referred to as the PRICE principle. This is an acronym for **protection, rest, ice, compression, and elevation**. The goal of this treatment is to: Reduce pain and swelling

Protection

In this context, it means to remove the athlete from play to protect against further injury. This is especially important in the first 48 hours after the injury occurred.

Rest

The athlete should not continue with any sporting activity following an ankle injury. There should be minimal load placed on the foot or ankle in the first 24 hours. Using crutches could be a good way to offload the foot and ankle.

Ice

The aim of applying ice is to relieve pain. 20 minutes with an ice pack every other hour for a day or two has a good effect.

Even though there are many commercial ice products available, the best solution is often a plastic bag filled with crushed ice and some water. Place a damp towel between the ice pack and skin.

Compression

Following an ankle injury, the most important thing is to apply a pressure bandage around the foot and ankle. Additional pressure can also be applied by placing a piece of foam (or rolled up tissues, etc.) on both sides of the ankle.

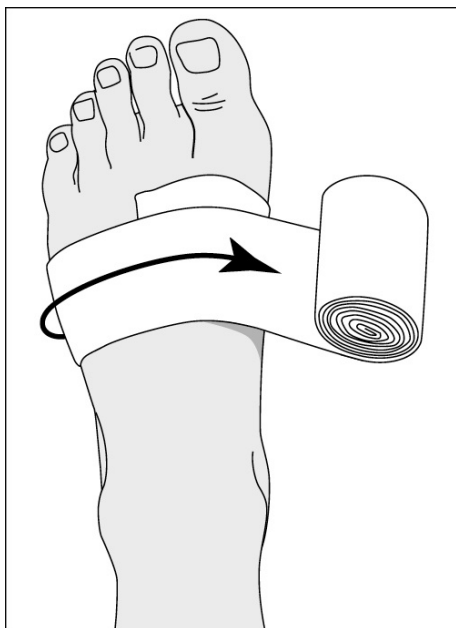
Applying compression minimizes swelling, which in turn may decrease stiffness. Use an elastic bandage and roll it out in a herringbone pattern (see video above). Start all the way out at the toes and work your way up to just above the injured area. The bandage should be as taut as possible without cutting off blood circulation. The use of a pressure bandage should be continued for the first 2-3 days.

Elevation

Swelling can be reduced further by keeping the leg elevated, preferably with the foot above heart level. This is particularly important in the first few hours, but it is best to continue to keep it elevated as much as possible for the first 24 hours. Remember that compression should be maintained around the clock to keep internal bleeding (swelling) to a minimum.

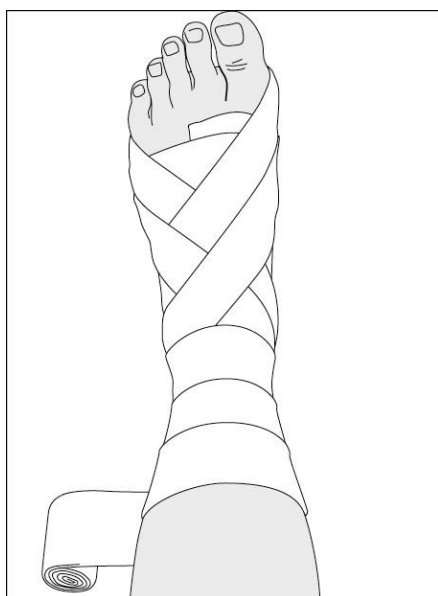
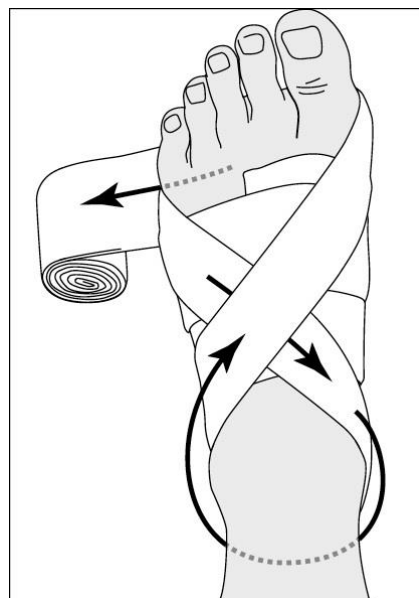
How to Strap an Ankle

1. Open up the elastic bandage.
2. Start at the toes, where they join the rest of the foot; tips of the toes are left unwrapped.
3. Start bandaging the foot by wrapping the bandage several times around the body of the foot.
4. Each new wrap of the bandage should overlap the previous wrap by approximately half as you work your way up the foot.
5. At the ankle, wrap the bandage in a figure-eight around the back of the ankle.
6. After strapping the ankle itself, work up the calf as far as the bandage will reach to provide additional support.
7. The bandage needs to be applied firmly to provide support but shouldn't be so tight as to constrict blood flow to the limb.



Starting where the toes join the rest of the foot, bandage the foot by wrapping the bandage several times around the body of the foot. Each new wrap of the bandage should overlap the previous wrap by approximately half as you work your way up the foot.

At the ankle, wrap the bandage in a figure-eight motion around the back of the ankle.



After strapping the ankle itself, work up the calf as far as the bandage will reach to provide additional support.

Burns due to fire and boiling water

Call for help if:

- The burn penetrates all layers of the skin
- The skin is leathery or charred looking, with white, brown, or black patches.
- The person is an infant or a senior.



For All Burns

1. Stop Burning Immediately

- Put out fire or stop the person's contact with hot liquid, steam, or other material.
- Help the person "stop, drop, and roll" to smother flames.
- Remove smoldering material from the person.
- Remove hot or burned clothing. If clothing sticks to skin, cut or tear around it.

2. Remove Constrictive Clothing Immediately

- Take off jewelry, belts, and tight clothing. Burns can swell quickly.

Then take the following steps:

For First-Degree Burns (Affecting Top Layer of Skin)

1. Cool Burn

- Hold burned skin under cool (not cold) running water or immerse in cool water until the pain subsides.
- Use compresses if running water isn't available.

2. Protect Burn

- Cover with sterile, non-adhesive bandage or clean cloth.
- Do not apply butter, oil, lotions, or creams (especially if they contain fragrance). Apply a petroleum-based ointment two to three times per day.

3. Treat Pain

- Doctors can Give over-the-counter pain reliever such as acetaminophen (Panadol, Tylenol), ibuprofen (Advil, Motrin, Nuprin), or naproxen (Aleve, Naprosyn).

4. When to See a Doctor

Seek medical help if:

- You see signs of infection, like increased pain, redness, swelling, fever, or oozing.
- The person needs tetanus or booster shot, depending on date of last injection. Tetanus booster should be given every 10 years.
- The burn blister is larger than two inches or oozes.
- Redness and pain last more than a few hours.
- The pain gets worse.
- The hands, feet, face, or genitals are burned.

For Second-Degree Burns (Affecting Top 2 Layers of Skin)

1. Cool Burn

- Immerse in cool water for 10 or 15 minutes.
- Use compresses if running water isn't available.
- Don't apply ice. It can lower body temperature and cause further pain and damage.
- Don't break blisters or apply butter or ointments, which can cause infection.

2. Protect Burn

- Cover loosely with sterile, nonstick bandage and secure in place with gauze or tape.

3. Prevent Shock

Unless the person has a head, neck, or leg injury, or it would cause discomfort:

- Lay the person flat.
- Elevate feet about 12 inches.
- Elevate burn area above heart level, if possible.
- Cover the person with a coat or blanket.

4. See a Doctor

- The doctor can test burn severity, prescribe antibiotics and pain medications, and administer a tetanus shot, if needed.

For Third-Degree Burns

1. Call for help

2. Protect Burn Area

- Cover loosely with sterile, nonstick bandage or, for large areas, a sheet or other material that won't leave lint in wound.
- Separate burned toes and fingers with dry, sterile dressings.
- Do not soak the burn in water or apply ointments or butter, which can cause infection.

3. Prevent Shock

Unless the person has a head, neck, or leg injury or it would cause discomfort:

- Lay the person flat.
- Elevate feet about 12 inches.
- Elevate burn area above heart level, if possible.
- Cover the person with a coat or blanket.
- For an airway burn, do not place a pillow under the person's head when the person is lying down. This can close the airway.
- Have a person with a facial burn sit up.
- Check pulse and breathing to monitor for shock until emergency help arrives.

4. See a Doctor

- Doctors will give oxygen and fluid, if needed, and treat the burn.

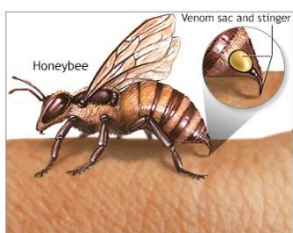
First aid on insects and honey bee bites

Many insects have the ability to bite and sting. Wasp and honey bee stings are the most common type of insect sting, as insects can be aggressive and sting without much provocation. Bees and certain types of ants can also bite and sting. The majority of insect stings cause a local reaction around the sting site. Often the area is painful, red, and swollen but improves after several days. Some people may develop a serious life-threatening allergic reaction to insect stings; this is known as anaphylaxis and can be fatal within minutes if left untreated.



Most minor insect stings will heal within a couple of days. In some cases the sting site may become infected and require treatment with antibiotics. Be vigilant for the signs of infection and seek medical advice if concerned.

First Aid Treatment for an Insect Sting



1. Look for an embedded stinger in the wound. If you can see a stinger, attempt to carefully remove it by brushing it away along the same angle as it entered the skin. Try to avoid using tweezers, as this can compress the stinger and squeeze more venom into the wound.
2. Clean the area with running water to reduce the risk of infection.
3. Elevate the area if a limb is affected and apply ice to reduce the swelling.
4. Medication such as antihistamines can reduce the itching from an insect sting.
5. Monitor the area for signs of infection developing.
6. Call EMS if the victim becomes unwell or displays any signs of developing a serious allergic reaction.

Watch Out for Infection

Insect stings are at risk of becoming infected. Watch out for these warning signs that may indicate an infected insect bite:

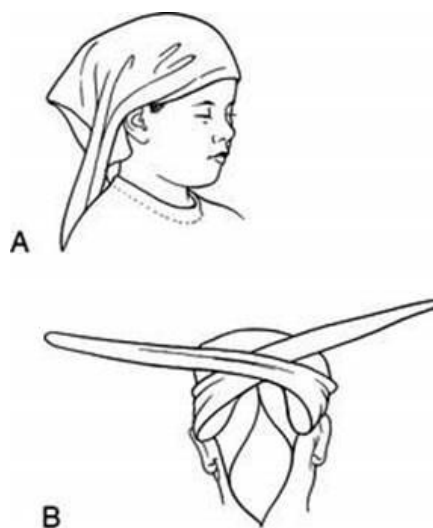
- Increasing pain
- A rapidly spreading red area around the wound
- Swelling
- Pus discharging from the wound
- High temperature (fever)
- Swollen glands near the wound

Do not delay seeking medical attention if you are concerned an insect bite is infected. If left untreated, the infection could spread to the bloodstream and cause blood poisoning (septicemia).

Covering head, hand, foot, and knee using Triangular Bandage

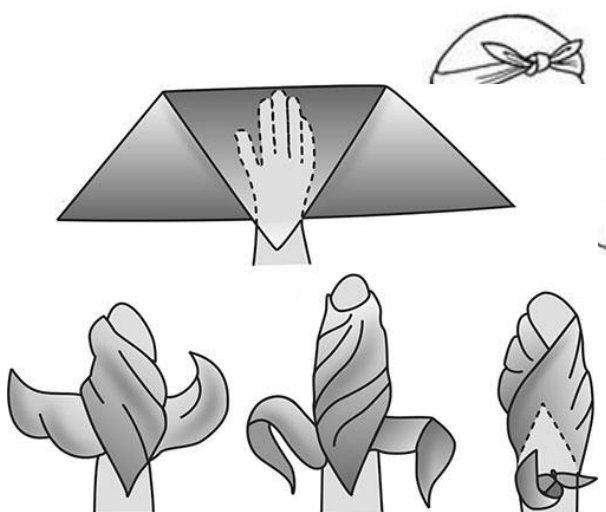
Head bandage

Bandaging the scalp is only modestly more challenging than what we've already done. Begin by folding the long edge over itself three times, about an inch or an inch and a half wide. Having done that, drape the bandage over the scalp, with the right angle going down the back of the neck and the middle of the folded part over the forehead. Bring the two ends of the folded part behind the head and to the middle of the scalp. To minimize the risk of this slipping up, put the folded part under the occiput—that's the bump you can feel on the back of your skull. Tie a reef knot in the middle of the forehead to anchor the bandage. Then, take the right angle and fold it up, tightening the part over the scalp then tucking the end into the folded part. If possible, and pinning it in place with a safety pin. It is much clearer if you look at the diagram



The Hand

Start by laying out the triangle bandage with the long side toward the person to be bandaged. Place the wounded hand, usually with the hand on top of the gauze, so that the edge of the triangle bandage is a about one to two inches past the wrist toward the elbow. The long finger of the hand should point to the right angle of the bandage. The diagram will make this clearer.

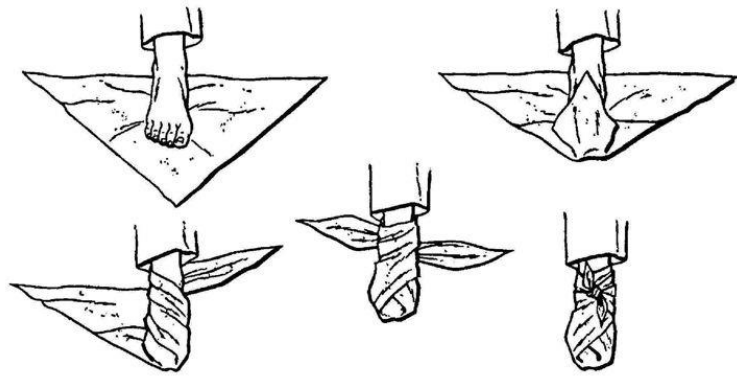


Bring the point of the right triangle back over the hand, then tuck the sides, clear to the points of the triangle, which will narrow the two bits of cloth going to the sides; again, the diagram should clarify this. With that done, wrap one side over the hand and under the wrist, being careful to keep a bit of pressure on the gauze without disturbing it. Do the same thing with the other one, gently tighten things up and then tie the two loose ends over the wrist with a square/reef knot. If there is a bit of the bandage sticking out from under the knot, it can be wrapped over the knot and tucked under it.

The Foot

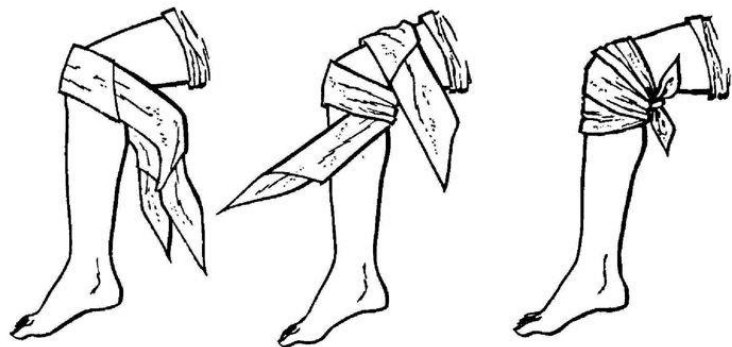
First, the foot lay on a triangle bandage on the ground, planting the injured foot on it so that the middle toe is pointed toward the right angle of the bandage, leaving enough bandage behind the heel to come up a little bit on the ankle. Fold the right angle of the bandage over the top of the foot, tucking the side of the bandage under the sole of the foot a

little bit to take up the extra material around the foot. Bring one of the corners of the bandage over the foot and behind the ankle, and then bring the other end around the other side and behind the ankle. Doing this will cause the part of the bandage behind the heel to come up over the heel; if it does not, lift it up and lock it in place. Then, anchor the bandage by tying a square knot on the front of the ankle. The figure below shows how this is done clearly.



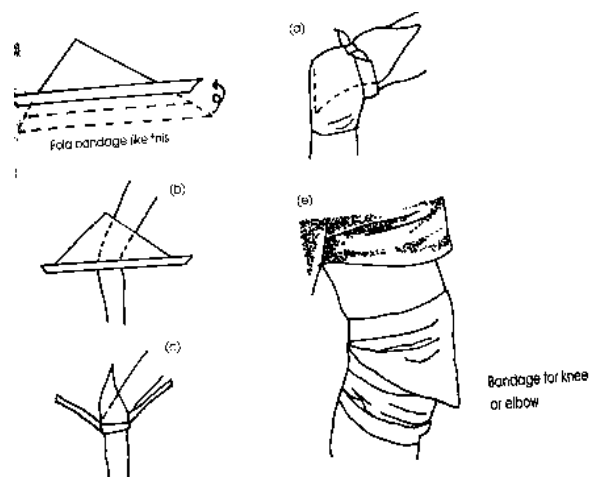
The Knee

Having taken care of the foot, let's move to the knee. Step one is to fold the triangle bandage into a cravat. With that done, put the middle of the cravat on the kneecap. Take one end of the cravat behind the knee then around the lower part of the knee. Take the other end, come around behind the knee and in front of the lower leg, right under the kneecap. Lock the bandage in place by tying a square knot behind the knee. Here is a simple diagram:



Second method

- Fold triangular bandage once or a twice at the edge as shown in figure
- Drape a triangular bandage below the knee.
- Create a hem and cover up the knee, tying with a reef knot at upper part
- Fashion a reef knot in the front. Tuck the tail that remains behind the head into the half knot



Preparing sling

Making a Sling for a Broken Arm

Most first aid kits contain triangular-shaped bandages to make arm slings. If an arm is broken, a sling can help by reducing movement until the victim can seek medical assistance.

How to Make a Sling

1. Place the triangular bandage underneath the injured arm with the middle point of the triangle sitting beneath the victim's elbow (think "point to the joint" to remember this).
2. Place the top end of the triangular bandage over the victim's opposite shoulder.
3. Bring the bottom end of the sling up and over the forearm and tie to the side of the victim's neck with a reef knot.
4. Ensure the sling is fully supporting the elbow and wrist.
5. Secure the point of the triangle with tape or a safety pin. In the event a triangular bandage is not available, you can improvise a sling with a scarf or a coat. As long as the arm is supported, it doesn't matter what material is used to make the sling.



Arm sling, steps 1 and 2.



Arm sling, steps 3 and 4.



Arm sling, step 5.

Get to know about symptoms and their first aid of following

- **Eating poison**
- **Shock**
- **Heart Attack**
- **Choking**
- **Seizure**

Poisoning

Poisoning occurs when a person comes into contact with a harmful substance. Our homes contain many potential poisons that can be dangerous if they are swallowed (ingested) or come into contact with our skin or eyes. In this section, we will look at four common poisoning situations and their first aid treatments. Always seek medical advice when dealing with a poisoning situation.



Alcohol and Drugs

The excessive alcohol or recreational drug use can have a variety of harmful effects on the body. Acute alcohol or drug intoxication can be life-threatening. Also, being intoxicated increases the risk of accidental death from other causes (for example, falling from a height or drowning), as alcohol interferes with our ability to make decisions and weigh risks. Alcohol intoxication can cause a victim to lose consciousness.

Signs and Symptoms of Alcohol or Drug Poisoning

- Vomiting
- Behavioral changes
- Confusion and aggression
- Reduced level of consciousness

First Aid Treatment for Alcohol or Drug Poisoning

1. Ensure the scene is safe for you to approach; victims who are intoxicated may be unpredictable or aggressive.
2. If the victim is unconscious, call EMS immediately; ensure her airway is open, and place her into the recovery position to reduce the risk of vomit blocking her airway (The Recovery Position).
3. If the victim is conscious, try to ask what she has taken and how much.
4. Seek medical assistance and ensure the victim is not left alone.
5. If an opioid overdose is suspected, administer nasal naloxone if available.

Shock

You've probably heard the term *shock* used when people talk about an injured victim, but what exactly is shock, and why is it so important in first aid? ***Shock occurs when our body tissues do not receive an adequate supply of oxygen.***

Oxygen is vital to allow the tissues in our body to work properly. Without oxygen our cells cannot function and will start to die. This causes damage to the brain and other vital organs. Severe **blood loss** is one cause of shock. This makes sense since there is less blood available to carry oxygen through the body to the cells. Shock is life threatening and requires urgent medical intervention to replace the blood lost.

There are other causes of shock besides blood loss. Major **burn injuries** can also cause shock as burns cause fluid to leak into the injury, and therefore, there is less fluid flowing through the body for the rest of the cells. In addition, a victim suffering from anaphylaxis (a major allergic reaction) may also go into shock.

You need to be aware of the signs and symptoms of shock and constantly monitor a victim for the warning signs that may indicate shock. The shock we've described here is different from mental or psychological shock following a distressing event. People who witness a traumatic event are often described as being "in shock," but this is not the same as physical shock caused by excessive blood loss.

Signs and Symptoms of Shock

- Pale, cold, or clammy skin
- Increased pulse and respiratory rate
- Weak pulse that is difficult to find
- Vomiting
- Confusion
- Reduced level of consciousness (late sign)

First Aid Treatment for Shock

1. Immediately call EMS.
2. Find and treat the cause of the shock (for example, by stopping any serious bleeding).
3. Lay the victim down and elevate the legs six to twelve inches unless there is evidence of a leg or pelvic (hip) injury.
4. Keep the victim warm and do not give him anything to eat or drink.
5. Provide reassurance and monitor vital signs.



After strapping the ankle itself, work up the calf as far as the bandage will reach to provide

Your priority in first aid is to identify shock in a victim and stop it from worsening. You cannot replace the blood lost by a victim, but effective first aid can stop further blood loss and prevent the shock from worsening. In addition, rapid treatment of burns and anaphylaxis can also help prevent shock.

Heart Attack

The heart is a muscular pump responsible for moving blood through the body. Our hearts need an excellent blood supply to provide enough oxygen to keep the heart muscle pumping effectively. A heart attack occurs when the blood supply to the heart is interrupted, leading to part of the heart muscle being injured or dying. This is different from cardiac arrest.

Why does a heart attack happen? The blood vessels that supply the heart are called the coronary arteries. As we age, our coronary arteries can become lined with plaque, which contains cholesterol and other fatty substances that build up in the wall of the artery. Lifestyle factors including smoking, obesity, and an unhealthy diet all accelerate this process.

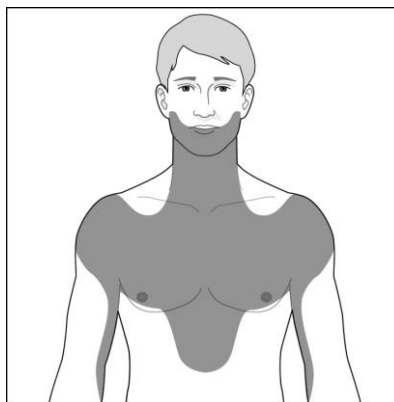
Over time, the fatty buildup in the lining of the artery increases. Eventually, this fatty buildup can rupture, resulting in the formation of a blood clot. This blood clot blocks the artery and stops blood from flowing to the vital heart muscle. The heart muscle becomes starved of oxygen and starts to die. This causes the victim to experience the common signs and symptoms of a heart attack.

A heart attack requires emergency medical treatment to unblock the artery and restore blood flow to the affected muscle. You must not delay in calling EMS if you suspect a victim is having a heart attack. The quicker the blockage in the artery is cleared, the more likely it is that the heart muscle will make a complete recovery.

Signs and Symptoms of a Heart Attack

- Chest pain that may radiate to the jaw, back, arms, or stomach
- Shortness of breath
- Excessive sweating
- Increased pulse and respiratory rate
- Pale skin
- Nausea and/or vomiting

The pain from a heart attack can vary. The classic description is central crushing chest pain. However, any sensation of pain or pressure in the chest should be considered as coming from the heart until proven otherwise.



The pain from a heart attack can spread into the jaw, arms, and stomach

Watch Out for Silent Heart Attacks

Some heart attacks may not present with the typical symptoms of chest pain radiating to the arm. In rare cases, a victim may have no pain at all. Women and people with diabetes are more likely to experience these “silent” heart attacks. Victims who have a silent heart attack may only have mild pain mistaken for heartburn or a pulled muscle. In some cases, the victim has no symptoms at all.

First Aid Treatment for a Heart Attack

If you suspect a victim may be having a heart attack, you should:

1. Immediately call EMS.
2. Place the victim in a comfortable position and loosen any tight clothing.
3. Offer to administer a regular strength (325 mg) aspirin tablet if appropriate.
4. Be prepared to perform cardiopulmonary resuscitation (CPR) if the victim loses consciousness and stops breathing.
5. Provide reassurance and monitor vital signs until emergency medical help arrives.

Administering Aspirin to a Heart Attack Victim

Early administration of aspirin has been shown to improve the chances of survival from a heart attack. Aspirin helps break down the clot in the blocked blood vessel.

- The aspirin tablet should be chewed and swallowed by the victim.
- Do not administer aspirin if the victim has a known aspirin allergy or a history of recent bleeding.

If you are concerned about administering aspirin, seek advice from EMS over the telephone or await the arrival of expert medical help. Most first aid kits contain emergency aspirin for use in a heart attack situation.

Choking

Nearly everyone will suffer an episode of choking in his or her lifetime. In the majority of cases it is mild, and a simple cough resolves the episode. However, choking can also be a life-threatening situation and without immediate First Aid.

Choking occurs when an object obstructs the airway and prevents air from flowing in and out of the lungs. If air (and therefore oxygen) cannot get into the lungs and therefore into the blood, it cannot be delivered to the tissues of the body, the most important being the brain and heart. If the heart does not receive an adequate supply of oxygen-rich blood, it will stop beating, leading to cardiac arrest. Choking is therefore a common cause of accidental death, especially in infants and the elderly. The most common object causing obstruction in this age group is food, but any small object can obstruct the airway. Children under the age of five are most at risk of choking as they have smaller airways and are at higher risk of swallowing objects they shouldn't.

Signs and Symptoms of Choking

- Clutching the throat or chest
- Difficulty in breathing
- Coughing
- Wheezing or grunting noises
- Red face initially, then turning pale or blue
- Reduced level of consciousness

If choking continues, the victim will become unconscious and stop breathing.

First Aid Treatment for Choking (Adult)

The first step in helping an adult choking victim is to establish whether there is complete or partial blockage of the airway. If the victim can speak and cough, then the blockage is only partial, as air is still able to move in and out of the lungs. If this is the case, then you should:

1. Encourage the victim to cough to dislodge the object.
2. Provide reassurance and monitor.
3. Loosen any tight clothing around the neck.
4. Call EMS if the symptoms do not quickly resolve.

If the victim is unable to speak or cough, then there is a complete obstruction of the airway. This is a life-threatening medical emergency. You should:

1. Immediately call EMS.
2. Deliver abdominal thrusts (Heimlich maneuver):
 - a. Stand behind the victim and explain what you are going to do.
 - b. Put your arms around his body, make a fist with one hand, and place this just above the victim's belly button.
 - c. Grasp this fist with your other hand and firmly pull inward and upward. If unable to deliver abdominal thrusts, then deliver chest thrusts or back blows.
3. Continue giving abdominal thrusts, chest thrusts, or back blows until emergency medical help arrives.
4. If the victim loses consciousness, assess whether he is breathing normally. If he is not breathing normally, immediately update EMS and commence cardiopulmonary resuscitation (CPR) until the arrival of medical help.

Anyone who has suffered a serious choking episode should be assessed by a medical professional. Abdominal thrusts carry a risk of internal organ damage and bleeding, and therefore, these victims need an urgent medical assessment.



How to give abdominal thrusts.

Chest Thrusts

If you are unable to perform abdominal thrusts or the victim is heavily pregnant or obese, perform chest thrusts instead. To perform chest thrusts:

1. Stand behind the victim and explain what you are going to do.
2. Reach around and place one fist on the center of the breastbone (sternum).
3. Grasp this fist with your other hand and firmly pull inward.
4. After each chest thrust, check to see if the object has been dislodged.



A heavily pregnant or obese victim may require chest thrusts instead of abdominal thrusts.

Back Blows

If you are unable to perform abdominal thrusts then back blows can be used to attempt to dislodge the object. To deliver back blows:

1. Lean the victim forward and explain what you are going to do.
2. Firmly hit him between the shoulder blades, using the heel of your hand.
3. After each back blow, check to see if the object has been dislodged.

Pediatric Choking

Kids love putting things they shouldn't in their mouths. It's their way of interacting with, exploring, and learning from the environment around them. Unfortunately, this makes them vulnerable to suffering from a choking episode. Food, too, can be a choking hazard for children; tragically, in a single country a child dies approximately every five days from choking on food. Children under the age of five are most at risk from accidental choking. All

parents and caregivers should know how to manage a choking emergency. Prompt first aid measures can dislodge the object and unblock the airway. Let's start by looking at the signs and symptoms of choking.

Signs and Symptoms of Pediatric Choking

- Clutching the throat or chest
- Appearing distressed or panicked
- Difficulty in breathing
- Coughing
- Wheezing or grunting noises
- Turning pale or blue
- Reduced level of consciousness

If choking continues, the victim will become unconscious and stop breathing.

First Aid for a Choking Baby (Ages 0–1)

The emergency treatment for a choking baby is different because you cannot deliver abdominal thrusts to a baby. Instead, you should give back blows and chest compressions to attempt to dislodge the object:

1. Immediately call EMS.
2. Lay the baby down on your thigh with her head supported downward. Deliver five back blows to the baby, using the palm of your hand in the center of the baby's back. Check to see if the object has been dislodged after each blow.
3. Turn the baby over on your thigh while supporting her head.
4. Deliver up to five chest compressions. Place two fingers in the center of the chest and push inward and upward.
5. Repeat the cycle of back blows and chest compressions until the object is dislodged.
6. If the baby loses consciousness, assess whether she is breathing normally. If she is not breathing normally, immediately update EMS and commence cardiopulmonary resuscitation (CPR) until the arrival of medical help.

Do not perform abdominal thrusts on a baby. Do not place your fingers blindly into the baby's mouth to attempt to remove the object as you could accidentally push the object farther down and worsen the choking situation. If you can easily see an object protruding from the mouth, you can attempt to carefully remove it. It can seem unnatural, but remember chest compressions and back blows need to be delivered with enough force to dislodge the object and save the baby's life.

First Aid for a Choking Child (Ages 1 to Puberty)

The emergency treatment for a child is the same as for an adult victim. The first step in helping a choking child is to establish whether there is complete or partial blockage of his airway. If the child can speak and cough then the blockage is only partial, as air is still able to move in and out of the lungs. If this is the case, then you should:

1. Encourage the child to cough to dislodge the object.
2. Provide reassurance and monitor the child.
3. Call EMS if the symptoms do not quickly resolve.

If the child is unable to speak or cough, then there is a complete obstruction of the airway. This is a life-threatening medical emergency requiring urgent first aid intervention. You should:

1. Immediately call EMS.
2. Deliver abdominal thrusts (Heimlich maneuver):
 - a. Stand or kneel behind the child.
 - b. Put your arms around his body, making a fist with one hand, and place this just above the victim's belly button.
 - c. Grasp this fist with your other hand and firmly pull inward and upward.
3. Continue giving abdominal thrusts until emergency medical help arrives or the object is dislodged.
4. If the child loses consciousness, assess whether he is breathing normally. If he is not breathing normally, immediately update EMS and commence cardiopulmonary resuscitation (CPR) until the arrival of medical help.

Any child who has suffered a serious choking episode should be assessed by a medical professional. Abdominal thrusts carry a risk of internal organ damage and bleeding, and therefore these victims need an urgent medical assessment.



The correct position to deliver back blows to a choking baby.



The correct position to deliver chest compressions to a choking baby.

Seizure/ Epilepsy

Epilepsy is a common condition of the brain in which a person tends to have recurrent unprovoked seizures.

About 70% of people with epilepsy gain control of their seizures with medication. People who continue to have seizures are more vulnerable to the potential risks associated with seizures, especially when seizures occur without warning and impair awareness.

Epilepsy, like other long-term conditions such as asthma or diabetes, comes with certain risks. If left unchecked these can become very serious.

Seizure-related risks are higher when people have poorly controlled seizures. Good seizure control is the first step in reducing seizure-related risks.

Seizures can sometimes lead to injuries or falls, and they can occasionally be more serious – even contributing to, or causing, death. Different types of seizures carry different risks.

Your risk level depends on the type of seizures you have, and your lifestyle. For instance, poorly controlled tonic-clonic seizures pose the highest safety risk, and if you take part in activities such as mountain climbing, this risk is increased.

Life is never risk-free, but taking positive action to reduce your seizures, thinking about risks specific to you and discussing seizure management with your doctor are a start to reducing some of your seizure-related risks.

Epileptic seizure first aid

If you are with someone having a tonic-clonic seizure (where the body stiffens, followed by general muscle jerking), try to:

- Stay calm and remain with the person.
- If they have food or fluid in their mouth, roll them onto their side immediately.
- Keep them safe and protect them from injury.
- Place something soft under their head and loosen any tight clothing.
- Reassure the person until they recover.
- Time the seizure, if you can.
- Gently roll the person onto their side for a recovery position after the jerking stops.

Do not put anything into their mouth or restrain or move the person, unless they are in danger.

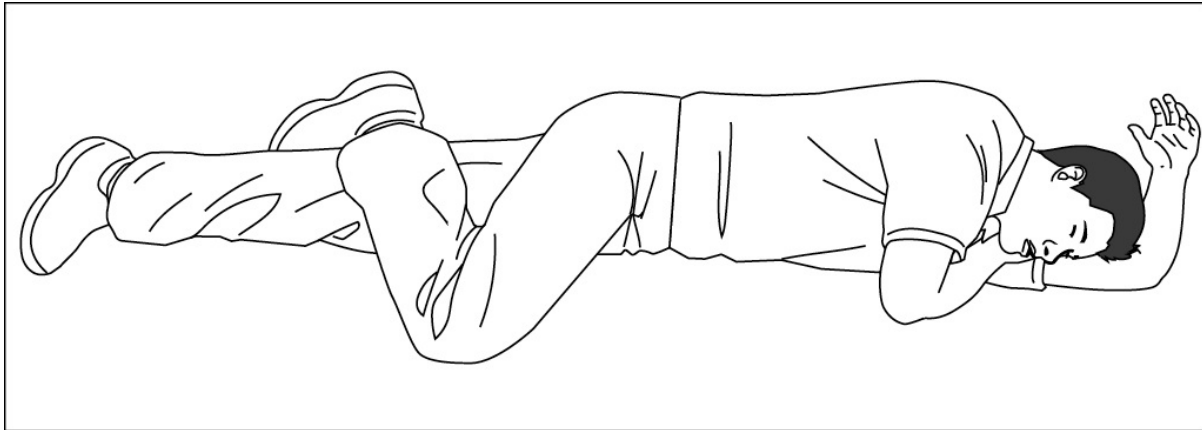
The Recovery Position

The recovery position, also known as the safe airway position, involves the victim being rolled onto his side in order to protect his airway and maintain effective breathing.

If you don't suspect a neck or a back injury then an unconscious but breathing victim should be placed in the recovery position to protect his airway. If you do suspect a neck or a back injury, then you should not move the victim unless you are unable to keep his airway open in the position in which he's found. Take a look at Neck and Back Injuries.

Why is the airway so important? A victim who is unconscious is at risk of blocking his airway as we discussed in the previous section. The tongue can fall back and obstruct the

airway, preventing air from flowing into the lungs. In addition, an unconscious victim is at risk of choking on his own vomit. At the entrance to the stomach is a muscle that prevents backflow of stomach contents up the gullet and into the throat. When you're unconscious, this muscle relaxes, and therefore stomach contents are free to flow up the gullet into the throat and then down into the airway and lungs. To reduce this risk with the help of gravity, an unconscious person should be rolled onto his side into the recovery position. In this position, any stomach contents will flow out of the victim's mouth and away from his airway.



The victim is on his side to reduce the risk of choking on his vomit.

There are different methods for placing a victim in the recovery position, and no one method is superior. The victim should be on his side with no pressure on the chest wall to obstruct his breathing.

Rolling a Victim into the Recovery Position

This version of the recovery position minimizes movement of the head and neck in case of a potential neck injury.

1. Kneel by the side of the victim.
2. Lift the victim's arm that is farthest away from you and place it above the victim's head with the palm facing upward toward the sky.
3. Place the victim's arm closest to you by his side.
4. Lift the victim's knee closest to you and place his foot flat on the floor.
5. Slide your arm that is closest to the victim's head underneath his shoulder blade and with that hand support his neck.
6. Roll the victim away from you by pushing on his shoulder; use your forearm with your hand supporting his neck, and use your free hand to push on his knee. Do not roll the victim by applying pressure to the head or neck.
7. As the victim rolls over, the head is cushioned by rolling onto his outstretched arm with your hand supporting his neck.
8. Ensure the airway is open and no pressure is on the chest wall.
9. Recheck victim's breathing to ensure he is still breathing normally.

Demonstration of CPR (Cardiopulmonary Resuscitation)

CARDIOPULMONARY RESUSCITATION (CPR) is one of the most well-recognized first aid techniques. However, did you know that the “**kiss of life**” is no longer required when performing CPR? First aid and CPR guidelines have changed dramatically over the past few years, and there is now a much greater emphasis on giving high-quality chest compressions. In this chapter, we’ll see describe the latest lifesaving first aid techniques to help a victim in cardiac arrest or during a choking emergency. The contents of this chapter could help you to save a life! We will start off by showing you how to **assess an unresponsive adult victim**, then how to perform CPR and use a defibrillator to help a victim of cardiac arrest. Finally, we’ll discuss choking and how to perform the **Heimlich maneuver** on a choking victim. We’ll be focusing on adult victims here; then for children and babies.

Assessing a Collapsed Victim

If you encounter a victim who has collapsed, you will need to act quickly to assess the situation and provide the appropriate First Aid measures. A useful way to remember how to approach this situation is by using the **DR ABC** action plan.

This action plan gives a structured way for you to assess the collapsed victim and remember what emergency action to take.

DR ABC Action Plan

DR ABC stands for:

- Danger
- Response
- Airway
- Breathing
- CPR

Let’s take a closer look at the individual steps of the DR ABC action plan and find out how you could save a victim’s life.

Danger

The first step (Danger) is to check for any potential hazards that could pose a risk to yourself, bystanders, or the victim. Quickly assess the scene and ensure it is safe for you to approach. If you identify any hazards, take steps to reduce the risk of harm to yourself or bystanders.

If available, wear disposable gloves in case of exposure to blood or other bodily fluids. Remember, you are the most important person in any emergency situation, and you are of no help if you become a second victim!

Response

Next, check for a response from the victim. Shout in both of her ears and firmly tap her shoulders. If you do not receive a response, then the victim is unconscious (see sidebar, What Is Unconsciousness?). This is a medical emergency, and at this point, you should call EMS if this has not been done.

What Is Unconsciousness?

If someone does not wake up when stimulated, then he is unconscious. An unconscious victim does not have an awareness of his surroundings or the events going on around him. This state is similar to being asleep, but the difference is that you can wake a person from sleep whereas an unconscious victim will not wake up.

Common Causes of Unconsciousness

- Cardiac arrest
- Severe head injury
- Low blood sugar levels
- Stroke
- Poisoning or intoxication
- Seizures
- Shock
- Hypothermia or heatstroke

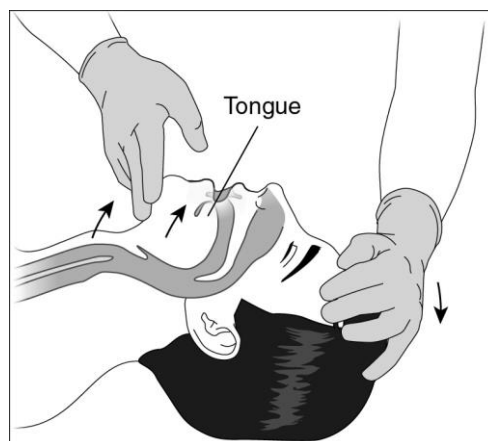
Airway

The next step in the DR ABC action plan is to open the victim's airway. An unconscious victim is at risk of blocking his airway due to the tongue falling back and preventing air from flowing in and out of the lungs. Let's take a look at why this occurs and what you can do about it in First Aid.

The airway refers to tubes and structures that air has to pass through to reach the lungs. The airway is at risk of becoming blocked if a victim is unconscious.

When a victim loses consciousness, his tongue becomes very floppy, falls backward in his throat, and can block his airway. When this happens, no air can reach the lungs. This situation can quickly result in death.

To open the airway, place one hand on the forehead and tilt the head backward. Place your other hand underneath the bony part of the chin and lift the chin upward. This maneuver is known as a *head tilt chin lift* and will move the tongue away from the back of the airway, enabling the victim to breathe.



The head tilt chin lift moves the tongue away from the airway at the back of the throat.

You suspect the victim may have a neck or a back injury, then the jaw thrust maneuver should be used instead. In this technique, the head is not tilted backward, so there is less movement of the victim's neck.

Breathing

After opening the victim's airway, check for regular normal breathing for a maximum of ten seconds. To do this, place your cheek just above the victim's mouth and look at his chest. Feel for exhaled air on the side of your cheek, look for the chest rising and falling, and listen for the sounds of breathing. The occasional, infrequent gasp is not normal breathing and should be treated as no breathing. Infrequent gasping is known as agonal breathing and occurs in victims immediately following cardiac arrest (when the heart stops beating).

CPR

Now that you've assessed for danger and victim response, opened the airway, and checked the victim's breathing, there are two different situations for you to consider. The most important decision you need to make is whether to commence cardiopulmonary resuscitation (CPR). This is the last step in the DR ABC action plan.

The Victim Is Not Breathing Normally

If the unconscious victim is not breathing normally, then immediately commence CPR. It is vital that the EME operator be kept informed of the victim's condition. Emergency services must be notified if the victim is not breathing so they can arrange an appropriate emergency medical response.

The Victim Is Breathing Normally

If the victim is breathing normally, perform a quick check to see if there is any major life-threatening bleeding. If you don't suspect a neck or a back injury, the victim should be rolled into the recovery position to further protect his airway.

Should I Check for a Pulse?

Studies have found that, in an unconscious patient who is not breathing, checking for a pulse is frequently inaccurate. In such a stressful situation you can often feel your own pulse through your fingers and mistake this for the victim's pulse. Additionally, you could be spending a long time trying to locate a pulse when actually it's not there or too weak to feel. Therefore, checking for the presence of a pulse during an initial assessment of a victim who is unconscious and not breathing normally is no longer recommended.

How to Perform CPR

Let's look at how to perform CPR on a victim of sudden cardiac arrest. We have already assessed danger, response, airway, and breathing, and established that the victim is not breathing normally. You must ensure that EMS has been called and then begin chest compressions.

For untrained bystanders, the Heart Association now recommends performing chest compressions only and not stopping to attempt rescue breaths.

High-quality chest compressions are the most important part of CPR, and it is vital that they are started quickly.

How to Give Effective Chest Compressions

To deliver high-quality chest compressions:

1. Kneel at the victim's side, parallel to his chest.
2. Place the heel of one hand in the middle of the victim's chest over the breastbone.
3. Place your other hand on top so your hands are parallel.
4. Push down to a depth of at least two inches (five centimeters).
5. Fully release and let the chest wall recoil.
6. Repeat and aim for 100 to 120 chest compressions per minute (maximum of two chest compressions per second).

Keep your elbows locked and use your upper body weight to deliver effective chest compressions. Ensure that you let the chest wall recoil fully after each compression and do not lean on the victim's chest.

The delivery of effective chest compressions requires a significant amount of energy from the rescuer. Evidence shows that chest compression quality decreases after only one to two minutes of CPR. If more than one rescuer is available, trade performing CPR every one to two minutes.



The proper technique for delivering chest compressions to a victim of sudden cardiac arrest.

Don't Worry About Rib Fractures

It is common for effective chest compressions to cause fractures of the victim's ribs and sometimes even his breastbone (sternum). Around a third of victims who receive CPR will sustain at least one rib fracture. If you feel rib fractures when performing CPR, recheck your hand position and the depth of chest compressions and continue performing CPR until EMS arrives. It can be off-putting if you hear or feel a rib fracture during CPR, but remember, you could save the victim's life. Any rib fractures you cause can be treated at a later date, but only if the victim is alive!

How to Give Effective Rescue Breaths

If you are a trained rescuer and willing to deliver rescue breaths, then perform two rescue breaths after each set of thirty compressions.

Continue this cycle of thirty chest compressions to two rescue breaths.

1. Open the victim's airway by tilting the head backward.
2. Pinch the nose to prevent air leaking out.
3. Take a regular breath and make a seal over the victim's mouth, using a disposable resuscitation face shield if available.
4. Breathe into the victim's mouth for approximately one second; do not overinflate the victim's lungs as this could cause air to go into the stomach and the victim to vomit.
5. Deliver two rescue breath attempts in total, then immediately resume chest compressions.

If two trained rescuers are present, one can perform chest compressions, and the other deliver rescue breaths, still at a ratio of thirty chest compressions to two rescue breaths. This two-person technique minimizes any interruptions in chest compressions.



Pinch the victim's nose and tilt the head backward to deliver effective rescue breaths.

Continue providing CPR (either chest compressions only, or chest compressions with rescue breaths) until the arrival of a defibrillator or emergency medical help. If a defibrillator is available, attach it to the victim and follow the instructions. When EMS arrives, they may ask you to assist by continuing chest compressions while they perform advanced medical interventions.

The victim may vomit when you are performing CPR. If this happens, don't panic. It does not mean that you've done anything incorrectly. As we saw in the previous section, an unconscious victim can lose control of his stomach contents and may vomit. If the victim vomits when you are performing CPR, turn him onto his side to allow the vomit to drain away from the airway. Then turn him back and resume CPR as quickly as possible.

When to Stop CPR

CPR needs to be performed continuously for it to be effective. Stop CPR only if:

- The victim shows signs of life and is breathing normally.
- EMS arrives and asks you to stop.
- The environment becomes too unsafe for you to remain on scene, and you cannot safely move the victim.
- You become physically exhausted and unable to continue. Remember you are the most important person in any emergency situation. If you become exhausted and bystanders are present, show them how to perform chest compressions so they can take over.

Demonstration of CPR (Cardiopulmonary Resuscitation) on Child or Baby

IN THIS CHAPTER, we're going to focus on common illnesses and emergency situations involving babies and children. Many first aid techniques, such as those used to treat minor injuries, are the same for children and adults.

However, there are key differences in the lifesaving skills of CPR and managing choking, and you need to be aware of these differences. In addition, children can develop certain medical conditions (for example, croup) that are not seen in adults. If you look after children on a regular basis or you are a parent, you should sign up for an accredited pediatric First Aid and CPR class in your area or check with your local fire department, so you can have hands-on practice in these emergency First Aid techniques. You never know when you might need to use them! Let's start off by looking at how to communicate with and assess the unwell child in a First Aid situation.

Looking After an Unwell Child or Baby

Looking after an unwell child can be challenging, as every parent or guardian knows. Unwell or injured children are often frightened and irritable and may not cooperate with your instructions. Nonverbal children are difficult to assess, as they cannot describe the symptoms they are experiencing or the events prior to the incident. You should be aware that children can deteriorate rapidly from a serious illness, so it is always important to seek medical assistance if you are concerned about an unwell child.

Children are not simply small adults and they develop different medical problems than adults. For example, children are very unlikely to have a heart attack but they are very prone to developing problems with their breathing such as croup and bronchiolitis. In addition, you need to be aware that normal vital signs such as respiratory and pulse rates in a child and baby are much quicker than in an adult. Below listed the average normal ranges, normal Pulse and Respiratory Rates, although these may vary slightly from child to child.

Communicating with an Unwell Child

It is important to gain the trust of a child who is unwell. She will understandably be frightened. Try these techniques to build trust and communicate with an unwell child.

- Drop down to the child's level; don't tower over her.
- Use easy-to-understand words and phrases (for example, a child may understand "hurt" but not "pain").
- Remain calm and in control of the situation; children will pick up if you display fear or anxiety.
- Provide lots of positive reinforcement and praise to the child.
- If you need to examine her or apply a bandage, try acting it out on Mom or Dad first to show the child what's involved.

Normal Pulse and Respiratory Rates

Children and babies will have different normal vital signs than adults. These values are highly variable and will depend on the exact age of the child, her emotional state, and whether she is awake or asleep. As a rough guide, the average pulse and respiratory rates decrease as the child grows older and enters adulthood.

- Less than 1 year: Pulse 100–160; respiratory rate 30–40
- 1–2 years: Pulse 100–150; respiratory rate 25–35
- 2–5 years: Pulse 95–140; respiratory rate 20–25
- 5–12 years: Pulse 75–120; respiratory rate 18–25
- More than 12 years: As per adults

Recognizing an Unwell Child or Baby

A child or baby who is unwell may display some of the following features:

- Irritability or being inconsolable
- Reduced oral intake of fluid and food
- Reduced urine output; in babies, fewer wet diapers
- Quietness and withdrawal
- In babies, reduced muscle tone (floppy)
- Pale and mottled skin (patchy and irregular colors)
- Difficulty in breathing

You must not delay in seeking urgent medical help if you are concerned about an unwell child.

Assessing an Unresponsive Child or Baby

When assessing an unresponsive child or baby, use the DR ABC action plan in order to remember the correct first aid steps. We discussed the DR ABC plan for adults already; in this section, we will look at the modifications you need to make when assessing an unresponsive child or baby.

DR ABC Action Plan

- **Danger**
- **Response**
- **Airway**
- **Breathing**
- **CPR (if required)**

Danger

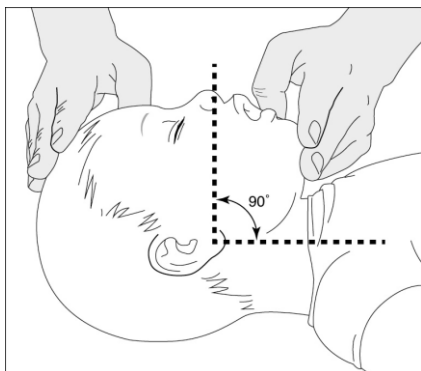
This remains the same as for an adult victim. Check for any hazards and ensure the scene is safe for you to approach before giving first aid. If the scene is too dangerous, stay back and call for EMS.

Response

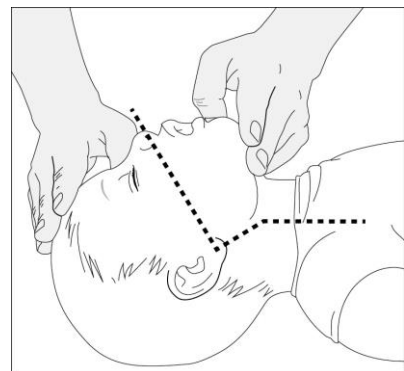
To check for a response in a child, shout in both ears and tap him on the shoulders. In a baby, shout and flick the bottom of his foot in order to gain a response. Never shake a child or baby. If a baby or child is unconscious, immediately call EMS if you have not already done so.

Airway

In children, open the airway using the “head tilt chin lift” maneuver. In babies, the procedure to open the airway is not the same as for adults and older children. The neck and throat of a baby are still developing, and the structures move differently than those of adults. The correct way to open the airway of a baby is to place the head in the neutral position as shown in the illustration. Do not tilt the baby’s head backward, as this will obstruct the airway. In addition, take care not to apply any pressure to the neck of the baby; this can also block the airway.



The correct position to open a baby's airway.



The incorrect position to open a baby's airway

Breathing

As with adults, check for the presence of normal breathing for up to ten seconds. To do this, place your cheek just above the victim’s mouth and look at his chest.

Feel for exhaled air on the side of your cheek, look for the chest rising and falling, and listen for the sounds of breathing. The occasional gasp is not normal breathing.

CPR

As with adults, the last step of the DR ABC action plan is deciding whether you should begin CPR.

The Child or Baby Is Not Breathing

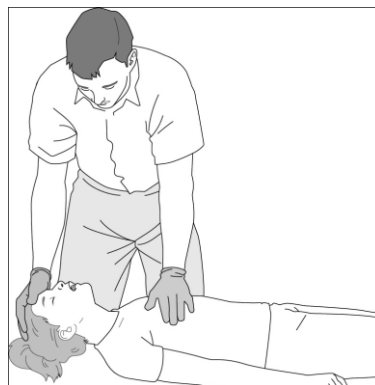
If the child or baby is not breathing normally, update EMS and immediately begin CPR. Let's see how to perform CPR on babies and small children.

The Child or Baby Is Breathing

If the child or baby is breathing normally but is unconscious, perform a quick check to see if there is any major life-threatening bleeding. If you don't suspect a neck or back injury, the child should be rolled into the recovery. Small babies can be cradled on their sides, with their heads slightly downward. This position keeps the airway open and protected until emergency medical help arrives.

Child CPR (Ages 1 to Puberty)

When performing CPR on a child, chest compressions require less force than on an adult. It is acceptable to use one hand when delivering compressions. However, a large child may require the two-handed technique to achieve effective chest compression depth. Deliver chest compressions at the same rate as for an adult victim (100 to 120 chest compressions per minute) and in the same location (center of the chest). The depth of chest compressions on a child should be approximately one third the depth of the chest.

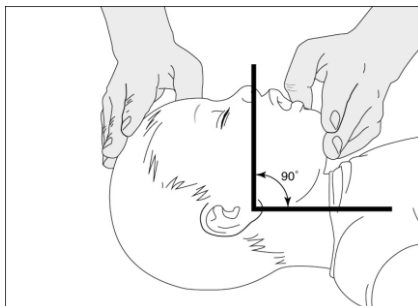


One-handed CPR can be used on a small child.

Rescue breaths for a child will require less force than for an adult victim. Ensure the child's airway is open and make a seal over her mouth. Use a disposable resuscitation face shield if available. Blow air in until the child's chest rises, then stop. Deliver two rescue breath attempts before immediately resuming chest compressions. If two rescuers are present, one can perform chest compressions and the other performs rescue breathing in order to minimize the interruption in chest compressions.

Baby CPR (Ages 0–1)

When performing CPR on a baby, chest compressions should be done by placing the pads of two fingers in the center of the baby's chest. Deliver chest compressions at the same rate as for an adult victim (100 to 120 chest compressions per minute). The depth of chest compressions on a baby should be approximately one third the depth of the chest.



Open the baby's airway by lifting the baby's chin and holding the head in the neutral position.

When performing rescue breaths on a baby, you may need to make a seal over the baby's mouth and nose. Take care not to extend the neck when performing rescue breathing, as this will close off the airway. Breathe in only until the baby's chest rises; this will require just a small puff of air. Do not overinflate a baby's lungs.

First aid on Snake bite and Scorpion Sting

Snakebites

Although many people have a fear of snakes, the chances of being bitten by a snake are very low. Luckily, fatalities from snakebites are rare, with only a handful reported each year in Pakistan. Snakes do not always inject venom when they bite, and when this occurs, it is called a dry bite. Although painful, dry bites do not come with the risks of the snake venom entering the body. Snake venom can cause a range of symptoms depending on the type of snake and the amount of venom that has entered the body. Identifying the snake involved can be very useful to help doctors guide the treatment given to the victim of a bite; however, do not risk being bitten by attempting to capture or kill the snake.



Signs and Symptoms of Snake Venom Toxicity

- Extreme pain and swelling around the bite
- Numbness around the affected area
- Muscle weakness
- Blurred vision
- Excessive drooling
- Vomiting
- Decreased level of consciousness

First Aid Treatment for Snakebite

1. Consider your own safety; do not attempt to capture or kill the snake.
2. Call EMS.
3. Keep the victim as still as possible with the bitten area below the level of the heart.
4. Wash the bite with running water if available and apply a sterile bandage over the bite.
5. If available, use an elastic bandage to lightly compress the tissues above the snakebite, working your way up the affected limb. Do not apply the bandage so tightly as to cut off blood flow. You should be able to fit two fingers underneath the bandage.
6. Make note of a description of the snake and hand over this information to EMS.
7. Provide reassurance and monitor vital signs.

Common Snakebite First Aid Myths

There are a number of myths regarding the correct first aid treatment for a snakebite. Unfortunately, these myths could make the situation worse and cause the victim further harm.

Do not:

- Apply a tight tourniquet to cut off blood supply.
- Cut or bite into the wound to remove the snake venom.
- Attempt to suck out the snake venom.
- Give the victim any alcohol or caffeine to drink.

These treatments are no longer recommended and may cause the victim more harm than good.

Scorpion Sting

The pain you feel after a scorpion sting is instantaneous and extreme. Any swelling and redness will usually appear within five minutes. More severe symptoms, if they're going to occur, will come on within the hour.

It's possible to die from a scorpion sting, though unlikely. There are an estimated

1,500 species of scorpion in the world, and only 30 of these produce venom toxic enough to be fatal. There is only one species of venomous scorpion, the bark scorpion.

Scorpions are predatory creatures that belong to the arachnid family. They have eight legs and can be recognized by their pair of grasping claws, which resemble pinchers, and their narrow, segmented tail. This tail is often carried in a forward curve over a scorpion's back and ends with a stinger



Symptoms and side effects of scorpion stings

The majority of scorpion stings only cause localized symptoms, such as warmth and pain at the site of the sting. Symptoms can be extremely intense, even if swelling or redness isn't visible.

Symptoms at the site of the sting can include:

- intense pain
- tingling and numbness around the sting
- swelling around the sting

Symptoms related to widespread effects of venom can include:

- breathing difficulties
- muscle thrashing or twitching
- unusual movements of the neck, head, and eyes
- dribbling or drooling
- sweating
- nausea
- vomiting
- high blood pressure
- accelerated heart rate or irregular heartbeat
- restlessness, excitability, or inconsolable crying

It's also possible for people who've been stung previously by scorpions to have an allergic reaction to a subsequent sting. It's occasionally severe enough to cause a life-threatening condition called anaphylaxis. Symptoms in these cases are similar to those of anaphylaxis caused by bee stings and can include trouble breathing, hives, nausea, and vomiting.

First Aid for Scorpion Stings

If a scorpion stings you or your friend, follow the suggestions below. Healthy adults may not need further treatment, and these tips can help keep children safe until they see a doctor:

- Clean the wound with mild soap and water.
- Apply a cool compress to the affected area. This may help reduce pain.
- Don't consume food or liquids if you're having difficulty swallowing.
- Take an over-the-counter pain reliever as needed.

How to avoid from stings?

To avoid stings:

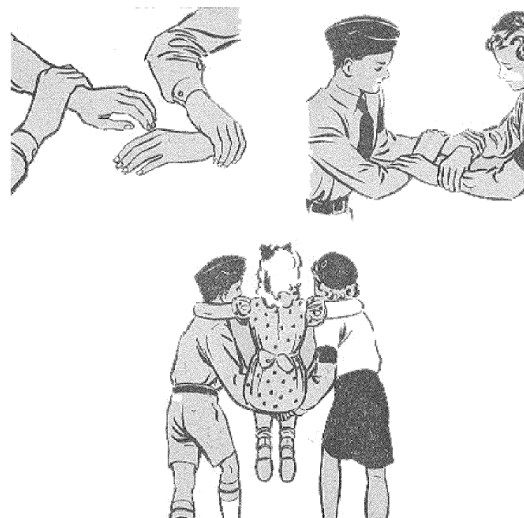
- Wear shoes, particularly at night.
- Put on gloves when you work in the yard, lift rocks and logs, or collect firewood.
- When you camp, don't sleep on the bare ground.
- Shake out your shoes before you put them on, especially if you've left them outside or in a basement or garage.

Using only hands for transporting patient

FOUR-HANDED SEAT

If a casualty is able to use their hands or arms to help hold themselves stably you can use the four-handed seat. A four-handed seat involves each First Aider to:

1. hold each opposite wrist of the other First Aider (left to right and right to left)
2. squat to a height that the casualty can easily sit on the hands
3. position the hand-seat midway down the casualty's thigh
4. place the casualty's arms around each First Aiders shoulders
5. steadily rise at the same time
6. Use crossover step to walk



THREE-HANDED SEAT

If a casualty is injured there lag and it need a support to hold them we use the three-handed seat. A three-handed seat involves each First Aider to:

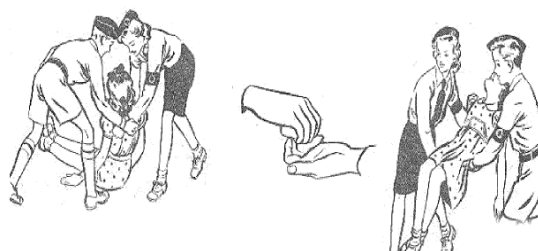
1. one must hold his wrist and grab a wrist of other First Aider (with one hand moving freely)
2. squat to a height that the casualty can easily sit on the hands and hold to the injured lag with free hand.
3. place the casualty's arms around each First Aiders shoulders
4. steadily rise at the same time
5. Use crossover step to walk



TWO-HANDED SEAT

A two-handed seat can be used for a conscious casualty who cannot put pressure on either leg, requires support from both First Aiders and can be comfortably carried whilst seated. A two-handed seat involves each First Aider to

1. hold the opposite wrist of the other First Aider (left to right or right to left)



2. squat to a height that the casualty can easily sit on the hands
3. steadily rise at the same time
4. Use crossover step to walk

First Aid knowledge about:

- Nervous Pressure
- Mental Stress
- Depression

Mental health includes our emotional, psychological, and social well-being. It affects how we think, feel, and act. It also helps determine how we handle stress, relate to others, and make choices. Mental health is important at every stage of life, from childhood and adolescence through adulthood.

Why is mental health important for overall health?

Mental and physical health is equally important components of overall health. For example, depression increases the risk for many types of physical health problems, particularly long-lasting conditions like diabetes, heart disease, and stroke. Similarly, the presence of chronic conditions can increase the risk for mental illness.

Over the course of your life, if you experience mental health problems, your thinking, mood, and behavior could be affected. Many factors contribute to mental health problems, including:

- Biological factors, such as genes or brain chemistry
- Life experiences, such as trauma or abuse
- Family history of mental health problems

Mental health problems are common but help is available. People with mental health problems can get better and many recover completely.

Early Warning Signs

Not sure if you or someone you know is living with mental health problems? Experiencing one or more of the following feelings or behaviors can be an early warning sign of a problem:

- Eating or sleeping too much or too little
- Pulling away from people and usual activities
- Having low or no energy
- Feeling numb or like nothing matters
- Having unexplained aches and pains
- Feeling helpless or hopeless
- Smoking, drinking, or using drugs more than usual
- Feeling unusually confused, forgetful, on edge, angry, upset, worried, or scared

- Yelling or fighting with family and friends
- Experiencing severe mood swings that cause problems in relationships
- Having persistent thoughts and memories you can't get out of your head
- Hearing voices or believing things that are not true
- Thinking of harming yourself or others
- Inability to perform daily tasks like taking care of your kids or getting to work or school

Don't ignore signs or symptoms that you notice in others, and don't assume they will go away. And remember not everyone will show the same signs and symptoms

First Aid for Mental Health

Approach the person

- Give the person a chance to talk to you. If they don't open up to you themselves, and you are concerned about them, initiate the conversation. Be open and honest about your concerns about their mental health.
- Make sure you approach the person in a comfortable space and at a time when you won't be interrupted. Make sure you're both sober and calm.
- Speak from your own perspective. Use 'I' statements, such as 'I have noticed...' and 'I feel concerned about...' rather than 'you' statements, such as 'You seem to be withdrawn...' or 'You're not eating and sleeping...'.
 - Say you're concerned and you're here to help.
 - Respect the person's own interpretation of their symptoms.
 - If the person doesn't want to talk to you, encourage them to talk to someone else they trust

Be supportive

- Use phrases that will help the person feel listened to, understood and hopeful. Some examples are 'I'm here for you', 'I can see this is a really hard time for you', and 'What can I do to help? Just tell me how'.
- Show the person dignity and respect.
- Don't blame.
- Be consistent in your emotional support and understanding.
- Encourage the person to talk to you or someone else.
- Listen well.
- Give the person hope for their recovery.
- Find accurate and appropriate resources for more information if the person wants it.

Know what's not helpful

Sometimes even with the best intentions, we can make matters worse. When reaching out to someone you suspect might have a mental health problem, avoid:

- telling them to 'snap out of it' or 'get over it'
- being hostile or sarcastic
- getting over-involved or over-protective
- nagging
- trivializing their experience (for example, don't tell them to smile or get their act together)
- belittling or dismissing their feelings
- being patronizing
- trying to cure them or solve their problems.

Encourage someone to seek help for a mental health problem

Ask the person if they need help to manage how they're feeling. If they want help, a good place to start is a visit to their GP. You can also chat to them about their options, particularly local and online services. Encourage them to act on their options.

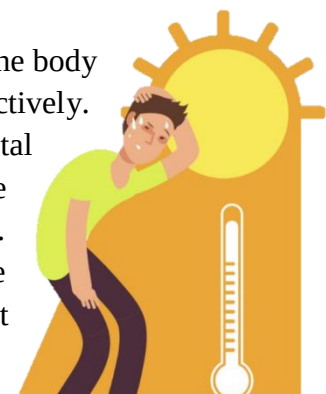
If the person doesn't want help, try to find out why. They may have some mistaken beliefs about getting help or their options. Try to help them feel better about seeking help.

If the person still resists help, tell them they can contact you if they change their mind. Respect their right not to seek help unless you believe they are at risk of harming themselves or others.

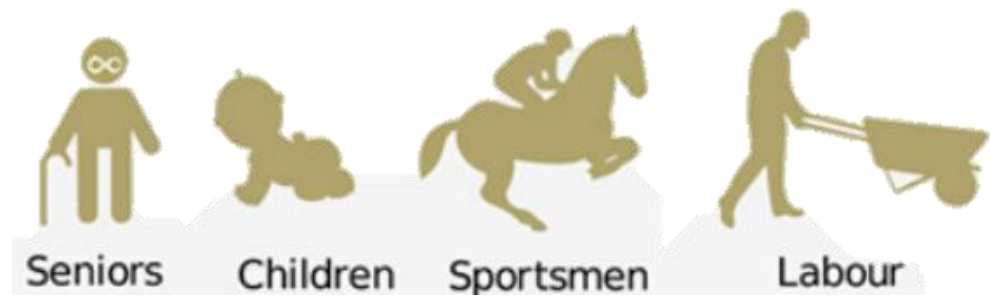
First aid for Heatstroke, Electric Shock, and in Unconsciousness

Heatstroke

Heatstroke is a life-threatening situation that occurs when the body becomes too hot and is unable to regulate temperature effectively. Heatstroke can be deadly, as the high body temperature causes vital organs to rapidly shut down. In addition, victims of heatstroke are often dehydrated because excessive fluid is lost through sweating. A victim of heatstroke has a high risk of dying unless immediate steps are taken to reduce his body temperature and restore lost fluid. Active cooling (for example, by using running water) is the main treatment in a heatstroke situation. All victims with suspected heatstroke need a thorough medical assessment, as they can develop complications affecting the heart, kidneys, and other vital organs. Children and the elderly are most at risk of developing heatstroke, so you need to be vigilant, especially during periods of hot weather. Heatstroke can also be brought on by exercising in very hot conditions, and athletes or

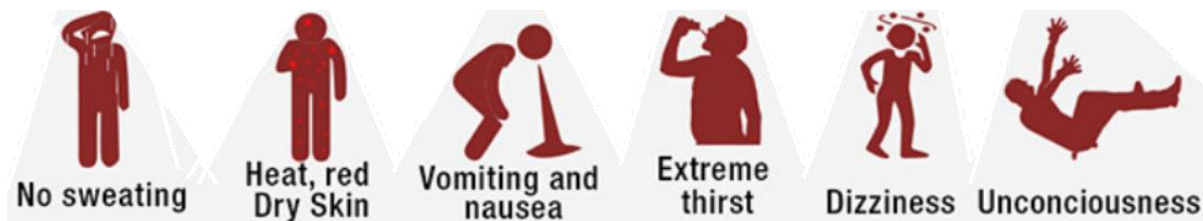


outdoor workers should be aware of the risk of heatstroke and know how to spot the warning signs in their peers.



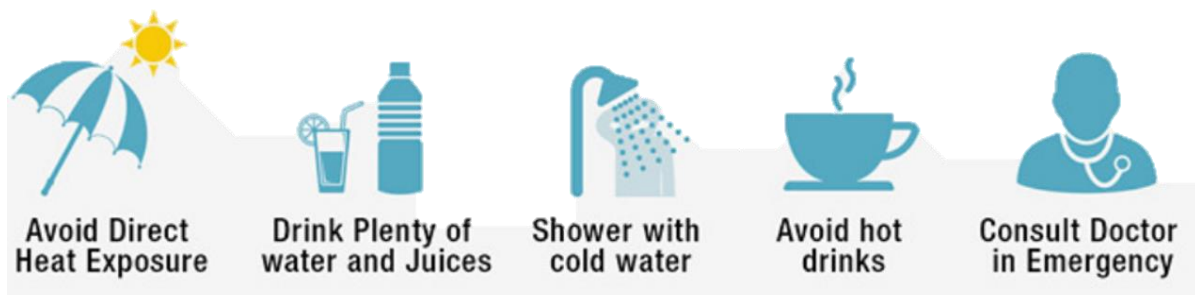
Signs and Symptoms of Heatstroke

- Hot and dry skin
- Body temperature above 105°F (41°C)
- Reduced level of consciousness and confusion
- Increased pulse and respiratory rate
- Extreme thirst



First Aid Treatment for Heatstroke

1. Immediately call EMS.
 2. Move the victim to a cool and sheltered place.
 3. Remove any excessive outer clothing.
 4. Cool the victim by using cool (but not freezing) running water and by placing ice-packs in his groin and armpits.
 5. If the victim is fully conscious and able to swallow, give him sips of cool water to drink.
 6. If the victim becomes unconscious, he will need to be turned into the recovery position to protect the airway.
 7. Be prepared to perform cardiopulmonary resuscitation (CPR) if the victim loses consciousness and stops breathing normally (see Cardiopulmonary Resuscitation (CPR)).
- Take care not to overcool the victim and cause hypothermia.



Electrical Shock

An electric shock happens when an electric current passes through your body. This can burn both internal and external tissue and cause organ damage.

A range of things can cause an electric shock, including:

- power lines
- lightning
- electric machinery
- electric weapons, such as Tasers
- household appliances
- electrical outlets

While shocks from household appliances are usually less severe, they can quickly become more serious if a child chews on an electric cord or puts their mouth on an outlet.

Aside from the source of the shock, several other factors affect how serious an electric shock is, including:

- voltage
- length of time in contact with the source
- overall health
- electricity's path through your body
- type of current (an alternating current is often more harmful than a direct current because it causes muscle spasms that make it harder to drop the source of electricity)

If you or someone else has been shocked, you may not need emergency treatment, but you should still see a doctor as soon as possible. Internal damage from electric shocks is often hard to detect without a thorough medical exam.

Symptoms of an Electric Shock

The symptoms of an electric shock depend on how severe it is.

Potential symptoms of an electric shock include:

- Loss of consciousness
- Muscle spasms
- Numbness or tingling
- Breathing problems
- Headache
- Problems with vision or hearing
- Burns
- Seizures
- Irregular heartbeat

Electric shocks can also cause compartment syndrome. This happens when muscle damage causes your limbs to swell. In turn, this can compress arteries, leading to serious health

problems. Compartment syndrome might not be noticeable immediately after the shock, so keep an eye on your arms and legs following a shock.

First Aid an Electric Shock

If you've been shocked

If you receive an electric shock, it might be difficult for you to do anything. But try to start with the following if you think you've been severely shocked:

- Let go of the electric source as soon as you can.
- If you can, call 1122 or local emergency services. If you can't, yell for someone else around you to call.
- Don't move, unless you need to move away from the electric source.

If the shock feels minor:

- See a doctor as soon as you can, even if you don't have any noticeable symptoms. Remember, some internal injuries are hard to detect at first.
- In the meantime, cover any burns with sterile gauze. Don't use adhesive bandages or anything else that might stick to the burn.



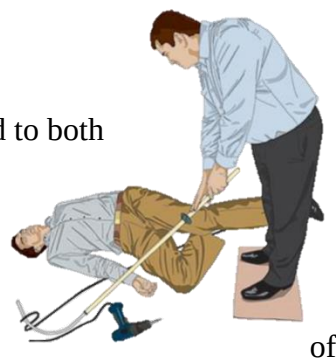
If someone else has been shocked

If someone else receives a shock, keep several things in mind to both help them and keep yourself safe:

- Don't touch someone who has been shocked if they're still in contact with the source of electricity.



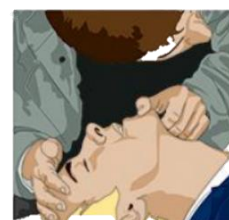
- Don't move someone who has been shocked, unless they're in danger of further shock.



- Turn off the flow of electricity if possible. If you can't, move the source of electricity away from the person using a non-conducting object. Wood and rubber are both good options. Just make sure you don't use anything that's wet or metal based.



- Stay at least 20 feet away if they've been shocked by high-voltage power lines that are still on.
- Call 1122 or local emergency services if the person was struck by lightning or if they came into contact with high-voltage electricity, such as power lines.
- Call 1122 or local emergency services if the person has trouble breathing, loses consciousness, has seizures, has muscle pain or



numbness, or is feeling symptoms of a heart issue, including a fast heartbeat.

- Check the person's breathing and pulse. If necessary, start CPR until emergency help arrives.
- If the person is showing signs of shock, such as vomiting or becoming faint or very pale, elevate their legs and feet slightly, unless this causes too much pain.
- Cover burns with sterile gauze if you can. Don't use Band-Aids or anything else that might stick to the burn.
- Keep the person warm.



Burns

Exposure to electricity can cause burns to the skin and, in severe cases, internal organs. In such cases the electricity may, for example, enter via a hand and leave via the feet causing 'entry' and 'exit' burns.

Responsive casualties

01. Burns should be immersed in cold running water for a minimum of 10 minutes (any constricting items such as watches should be removed). (Fig .1)
02. Once cooled, the burn should be covered with a sterile dressing (non-fluffy). (Fig .2)

Unresponsive casualties

Cool the burn with wet dressings after placing them in the recovery position.

Do not:

- + Burst any blisters.
- + Apply adhesive dressings.
- + Remove damaged skin.
- + Apply ointments or creams.
- + Cover with 'fluffy' dressings.
- + Affix dressing too tightly.
- + Apply butter, fats or margarine.
- + Remove damaged clothing.
- + Apply ice.



Unconsciousness

Unconsciousness is the state in which a person is unable to respond to stimuli and appears to be asleep. They may be unconscious for a few seconds — as in fainting — or for longer periods of time.

People who become unconscious don't respond to loud sounds or shaking. They may even stop breathing or their pulse may become faint.

This calls for immediate emergency attention. The sooner the person receives emergency first aid, the better their outlook will be.

Symptoms That A Person May Become Unconscious

That may indicate that unconsciousness is about to occur include:

- Sudden inability to respond

- Slurred speech
- A rapid heart rate
- Confusion
- Dizziness or lightheadedness

Administer first aid for Unconscious

If you see a person who has become unconscious, first check whether they're breathing.

If they're not breathing

If they're not breathing, have someone call 1122 or your local emergency services immediately and prepare to begin CPR.

If they're breathing

If they're breathing, take steps to get them into the recovery position. This helps them maintain a clear airway and decreases the risk of choking.

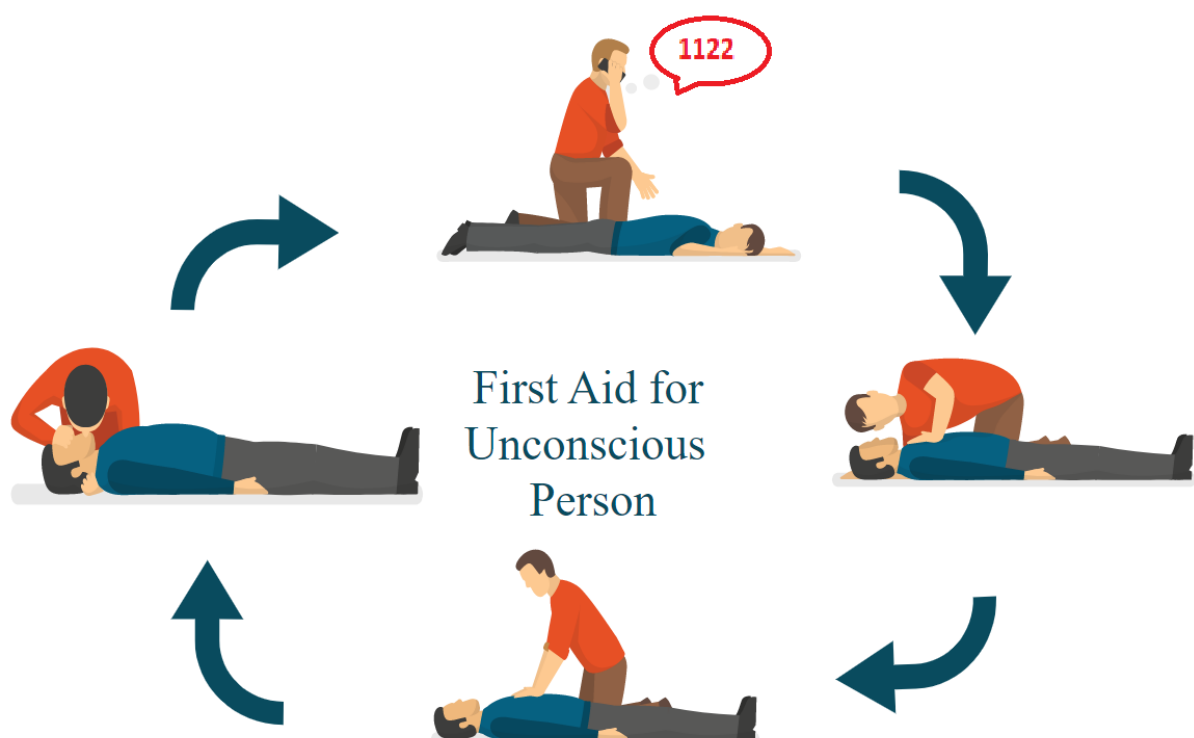
If they're bleeding heavily

If there's major bleeding occurring, find the source of the injury and place direct pressure on the bleeding area until the bleeding's slowed and expert help has arrived.

In cases where the person is bleeding from their limbs, you could also apply a tourniquet above the bleeding area until expert help arrives.

If the person has a severe wound, you should:

- Elevate the part of their body that's been injured (unless it's the head)
- Place a medium amount of pressure on their wound (unless they've injured their eye)
- Help them lie down (so that if they happen to faint, there's no chance of them falling and developing another injury)



Broken bones and its types along its first aid

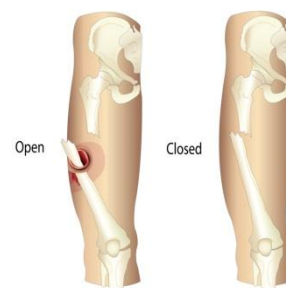
Fractures

When you break a bone, healthcare providers call it a bone fracture. A bone fracture is a medical condition that occurs when significant pressure is exerted on the bones, caused by: falls, traffic accidents, or bone stress (to which stress fractures in athletes are attributed). Besides, fractures may be attributed to some medical conditions that weaken the bones, such as: osteoporosis and some cancers. Fractures caused by diseases are referred to as pathologic fractures.

Common fractures are of two types:

Closed fracture: where the damaged bone does not tear through the skin; and

Open (compound) fracture: where the skin is torn and penetrated by the damaged bone. Open fractures are more serious.



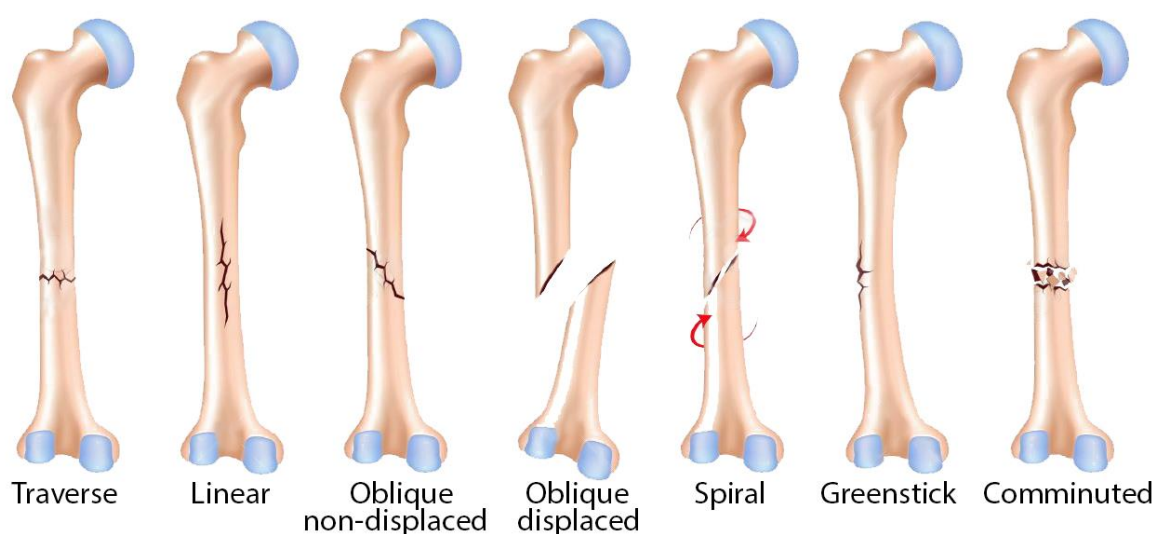
Healthcare providers can usually categorize a bone fracture based on its features. The categories include:

Complete fractures: The break goes completely through the bone, separating it in two.

Displaced fractures: A gap forms where the bone breaks. Often, this injury requires surgery to fix.

Partial fractures: The break doesn't go all the way through the bone.

Stress fractures: The bone gets a crack in it, which is sometimes tough to find with imaging.



A healthcare provider may add extra terms to describe partial, complete, open and closed fractures. These terms include:

Avulsion: A tendon or ligament pulls part of the bone off. Ligaments connect bones to other bones, while tendons anchor muscles to bones.

Comminuted: The bone shatters into several different pieces.

Compression: The bone gets crushed or flattened.

Impacted: Bones get driven together.

Oblique: The break goes diagonally across the bone.

Spiral: The fracture spirals around the bone.

Transverse: The break goes in a straight line across the bone.

First Aid for Fractures

Simple Fractures

1. Do not move the person unless necessary
2. Keep person warm and comfortable
3. Check affected area for swelling discoloration of skin
4. Do not move limb: stable with blanket or pillow
5. Seek medical help immediately or send person to get help

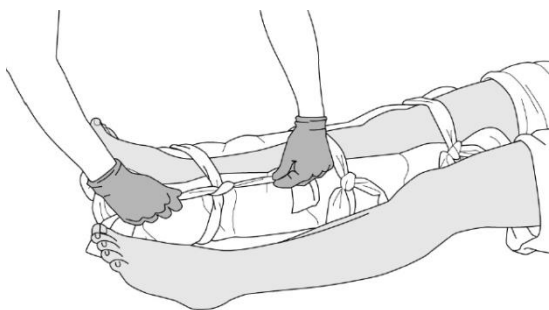
Compound Fractures

1. Do not move the person unless necessary
2. Do not allow person to eat or drink
3. Ensure person is breathing properly
4. Stop bleeding and test person of unconscious
5. Keep person warm and comfortable
6. Cover the injured part with a sterile pad
7. Apply a splint to keep the bone from causing further damage to immediate tissues

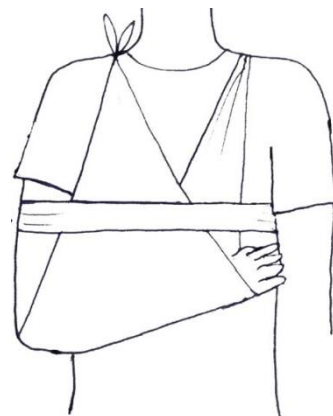
ARM: Bandage injured arm to body with sling

LEG: Bandage injured leg to uninjured one

8. Seek medical help immediately or send person to get help



For a fractured leg, pad around the injury.



For a fractured arm, sling

Respiratory system and two artificial means of giving respiration

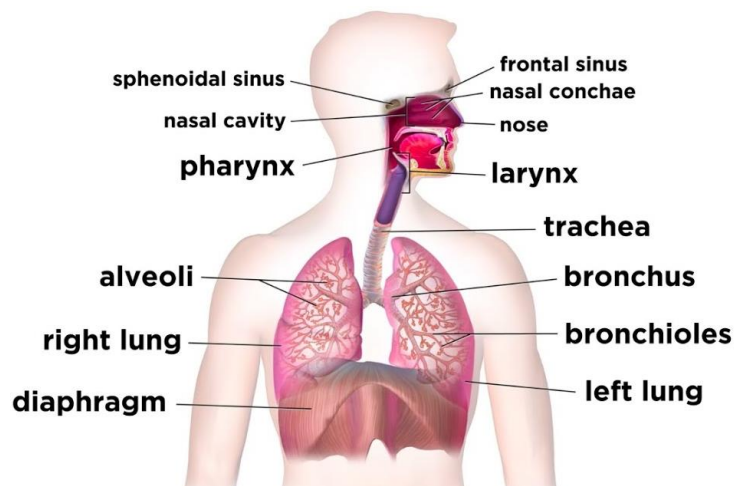
What Is the Respiratory System?

The respiratory system is the organs and other parts of your body involved in breathing, when you exchange oxygen and carbon dioxide.

Parts of the Respiratory System

Your respiratory system includes your:

- Nose and nasal cavity
- Sinuses
- Mouth
- Throat (pharynx)
- Voice box (larynx)
- Windpipe (trachea)
- Diaphragm
- Lungs
- Bronchial tubes/bronchi
- Bronchioles
- Air sacs (alveoli)
- Capillaries



Inhalation and Exhalation

Inhalation and exhalation are how your body brings in oxygen and gets rid of carbon dioxide. The process gets help from a large dome-shaped muscle under your lungs called the diaphragm.

When you breathe in, your diaphragm pulls downward, creating a vacuum that causes a rush of air into your lungs.

The opposite happens with exhalation: Your diaphragm relaxes upward, pushing on your lungs, allowing them to deflate.

Artificial means of giving respiration

Resuscitation by inducing artificial respiration consists chiefly of two actions:

- (1) Establishing and maintaining an open air passage from the upper respiratory tract (mouth, throat, and pharynx) to the lungs and
- (2) Exchanging air and carbon dioxide in the terminal air sacs of the lungs while the heart is still functioning. To be successful such efforts must be started as soon as possible and continued until the victim is again breathing.

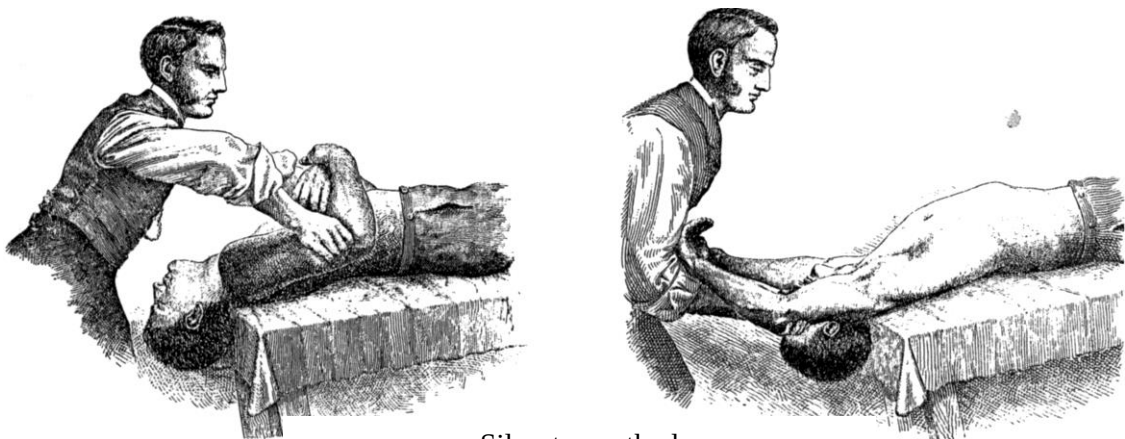
There are various methods of artificial respiration, most based on the application of external force to the lungs

1. **Mouth-to-mouth breathing** became the most widely used method of artificial respiration. The person using mouth-to-mouth breathing places the victim on his back, clears the mouth of foreign material and mucus, lifts the lower jaw forward and upward to open the air passage, places his own mouth over the victim's mouth in such a way as to establish a leak-proof seal, and clamps the nostrils. The rescuer then alternately breathes into the victim's mouth and lifts his own mouth away, permitting the victim to exhale. If the victim is a child, the rescuer may cover both the victim's mouth and the nose. The rescuer breathes 12 times each minute (15 times for a child and 20 for an infant) into the victim's mouth. If a victim was choking before falling unconscious



Head lift chin tilt method, and giving air blows

2. **In the Silvester method**, the victim was placed face up, and the shoulders were elevated to allow the head to drop backward. The rescuer knelt at the victim's head, facing him, grasped the victim's wrists, and crossed them over the victim's lower chest. The rescuer rocked forward, pressing on the victim's chest, and then backward, stretching the victim's arms outward and upward. The cycle was repeated about 12 times per minute.



Silvester method

Complications due to high blood pressure and low blood pressure

Blood pressure

Blood pressure (BP) is the pressure of circulating blood against the walls of blood vessels. Most of this pressure results from the heart pumping blood through the circulatory system. When used without qualification, the term "blood pressure" refers to the pressure in the large arteries. Blood pressure is usually expressed in terms of the systolic pressure (maximum pressure during one heartbeat) over diastolic pressure (minimum pressure between two heartbeats) in the cardiac cycle. It is measured in millimeters of mercury (mmHg) above the surrounding atmospheric pressure.



High blood pressure (hypertension)

High blood pressure (hypertension) is a common condition in which the long-term force of the blood against your artery walls is high enough that it may eventually cause health problems, such as heart disease.

Blood pressure is determined both by the amount of blood your heart pumps and the amount of resistance to blood flow in your arteries. The more blood your heart pumps and the narrower your arteries, the higher your blood pressure.

You can have high blood pressure for years without any symptoms. Uncontrolled high blood pressure increases your risk of serious health problems, including **heart attack and stroke**. Fortunately, high blood pressure can be easily detected. And once you know you have high blood pressure, you can work with your doctor to control it.

Complications

The excessive pressure on your artery walls caused by high blood pressure can damage your blood vessels as well as your organs. The higher your blood pressure and the longer it goes uncontrolled, the greater the damage.

Uncontrolled high blood pressure can lead to complications including:

- **Heart attack or stroke.** High blood pressure can cause hardening and thickening of the arteries (atherosclerosis), which can lead to a heart attack, stroke or other complications.
- **Aneurysm.** Increased blood pressure can cause your blood vessels to weaken and bulge, forming an aneurysm. If an aneurysm ruptures, it can be life-threatening.

- **Heart failure.** To pump blood against the higher pressure in your vessels, the heart has to work harder. This causes the walls of the heart's pumping chamber to thicken (left ventricular hypertrophy). Eventually, the thickened muscle may have a hard time pumping enough blood to meet your body's needs, which can lead to heart failure.
- **Weakened and narrowed blood vessels in your kidneys.** This can prevent these organs from functioning normally.
- **Thickened, narrowed or torn blood vessels in the eyes.** This can result in vision loss.
- **Metabolic syndrome.** This syndrome is a group of disorders of your body's metabolism, including increased waist size, high triglycerides, decreased high-density lipoprotein (HDL) cholesterol (the "good" cholesterol), high blood pressure and high insulin levels. These conditions make you more likely to develop diabetes, heart disease and stroke.
- **Trouble with memory or understanding.** Uncontrolled high blood pressure may also affect your ability to think, remember and learn. Trouble with memory or understanding concepts is more common in people with high blood pressure.
- **Dementia.** Narrowed or blocked arteries can limit blood flow to the brain, leading to a certain type of dementia (vascular dementia). A stroke that interrupts blood flow to the brain also can cause vascular dementia.

Low blood pressure (hypotension)

Low blood pressure is generally considered a blood pressure reading lower than 90 millimeters of mercury (mm Hg) for the top number (systolic) or 60 mm Hg for the bottom number (diastolic).

Complications

Potential complications of low blood pressure (hypotension) include:

- **Dizziness**
- **Weakness**
- **Fainting**
- **Injury from falls**

Severely low blood pressure can reduce the body's oxygen levels, which can lead to heart and brain damage

A task for a Boy Scout who is willing to have a First Aid Proficiency Badge

Get to know about Medical Conditions of your own friends

Each boy scouts must know about medical conditions about his friend, in case of emergency they must get to know, how to provide and apply a specific First Aid on his friend.

Please enlist on copy you made for a First Aid notes, a medical condition (if he had) of your friend and write down its First Aid procedure too.

Disclaimer

This book is intended to provide general information and guidance with regard to the treatment of acute injury or illness when professional medical attention is not readily available. The ideas, procedures, and suggestions in this book are intended to supplement, not replace, the advice of trained medical professionals. Medical attention should be immediately and urgently sought for any acute injury or illness of a serious or lasting nature. For any injury or illness of a chronic nature, please consult your physician before adopting any of the suggestions in this book. The author and publisher disclaim any liability arising directly or indirectly from the use of the information in this book.